

Wave–wave nonlinearities

15.1 Introduction

Wave nonlinearity is a vast subject and is not specific to plasma physics because nonlinear wave behavior occurs in virtually any physical medium where waves can propagate. However, because of the enormous variety of plasma waves, there is usually at least one plasma context where any given type of wave nonlinearity is an important issue. Three general types of nonlinear wave behavior will be discussed in this chapter: mode–mode coupling instabilities, self-modulation, and solitons. Before discussing these phenomena in detail, we first present a qualitative overview showing how certain basic wave nonlinearities are manifested.

Mode–mode coupling instabilities

Suppose a linear wave is excited in a plasma by an antenna driven by an appropriately tuned sine wave generator. In particular, suppose the plasma frequency is $\omega_{pe}/2\pi = 100$ MHz and the sine wave generator is tuned to a frequency well above the plasma frequency, say $\omega/2\pi = 500$ MHz, so as to cause the antenna to radiate an electromagnetic plasma wave with dispersion relation $\omega^2 = \omega_{pe}^2 + k^2 c^2$. This wave propagates through the plasma and is picked up by a distant receiving probe connected to a spectrum analyzer, a device that provides a graphic display of signal amplitude versus frequency. The received signal shows up on the spectrum analyzer display as a sharp peak at 500 MHz, as shown in Fig. 15.1(a). If the amplitude of the sine wave generator is increased or decreased, the peak on the spectrum analyzer display moves up or down proportionately, indicating the strength of the received signal.

An odd behavior is observed when the generator amplitude is increased above some critical threshold. At generator amplitudes below this threshold, the spectrum analyzer displays just the single peak at 500 MHz, but above this threshold, two additional peaks abruptly appear at two different frequencies and these new peaks have amplitudes considerably lower than the 500 MHz peak. These two additional

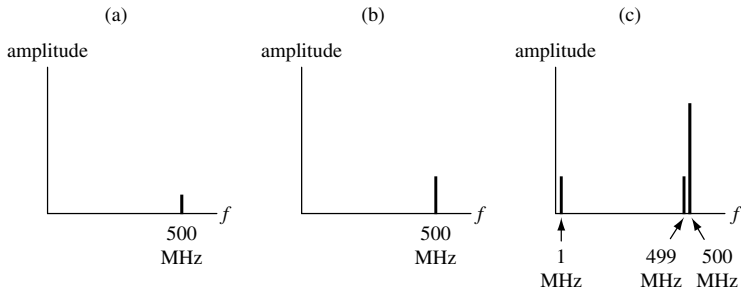


Fig. 15.1 Frequency spectra showing onset of decay instability at high power when a 500 MHz sine wave generator excites a wave in plasma. Low and intermediate generator power as in (a), (b) provides linear response, but high generator power as in (c) results in decay instability with daughter modes appearing at 1 MHz and 499 MHz.

peaks are typically at frequencies that sum to 500 MHz and the frequencies of the two additional peaks are typically very different from each other. For example, one of the two additional peaks might be at 1 MHz and the other at 499 MHz. These two peaks are called daughter waves and their spectrum is usually broader and their amplitude “springier” than the 500 MHz “pump” wave. If the generator amplitude is increased further, the pump wave amplitude no longer increases in proportion to the generator amplitude, and instead the daughter wave amplitudes increase. One gets the impression that the plasma does not like to have too much power in the pump wave and when the above-mentioned threshold is exceeded, excess power spills into the daughter waves at the expense of the pump wave. If the power is increased further, the daughters become quite large, and then at another threshold, the upper daughter might suddenly spawn its own pair of low- and high-frequency daughters. The spectrum becomes quite complicated and the amplitudes of the various spectral components depend sensitively on both the generator amplitude and on the plasma parameters. One gets the impression that energy is sloshing around between the pump wave and the daughter waves.

This behavior can occur at surprisingly modest generator power levels where it might have reasonably been expected that nonlinear behavior would be negligible. The springiness of the daughter wave amplitudes suggests some kind of resonant effect makes the nonlinearities more important than would be expected. The behavior sketched here is sometimes called a parametric decay instability (Silin 1965) because the pump wave is considered as decaying into the daughter waves. It has analogies to photon decay and in a sense can be considered as the classical limit of photon decay. Whether this sort of instability is good or bad depends

on the context. If the goal is to propagate a large amplitude wave through a plasma and the wave decays into daughters, then the instability would be bad and efforts would be required to avoid it. On the other hand, if one of the daughter waves is normally difficult to excite and has some beneficial aspects, then the decay instability provides a means to access the desired daughter wave. Another possibility is that the onset of the decay could be used as a diagnostic to provide information about the plasma.

Self-modulation

A closely related situation is where the low-frequency daughter wave is at zero frequency. In this case the pump wave beats with itself and so modulates the equilibrium via a so-called “ponderomotive” nonlinear pressure, which acts to expel plasma. If this happens, increasing the generator amplitude above the nonlinear threshold causes the pump wave to dig a hole in the plasma density. The pump wave can dig itself a channel in the plasma and then this channel can guide, focus, and even trap the pump wave.

Solitons

Plasma waves are typically dispersive so, after propagating some distance, an initially sharp pulse will become less sharp because the various components of the Fourier spectrum constituting the pulse propagate at different phase velocities. In certain situations, however, pulses with an amplitude exceeding some critical value will propagate indefinitely without broadening, even though the medium is nominally dispersive. This high-amplitude non-dispersive pulse is called a soliton and its existence results from nonlinearities competing against dispersion in a way such that the two effects cancel each other.

15.2 Manley–Rowe relations

Before investigating wave–wave nonlinearities, it is instructive to examine a closely related, but simpler, system consisting of three nonlinearly coupled harmonic oscillators (Manley and Rowe 1956). This simple system consists of a particle of mass m moving in a three-dimensional space with dynamics governed by the Hamiltonian

$$H = \frac{1}{2m} (P_1^2 + P_2^2 + P_3^2) + \frac{1}{2} (\kappa_1 Q_1^2 + \kappa_2 Q_2^2 + \kappa_3 Q_3^2) + \lambda Q_1 Q_2 Q_3. \quad (15.1)$$

In the $\lambda \rightarrow 0$ limit this Hamiltonian describes three independent harmonic oscillators having respective frequencies $\omega_1 = \sqrt{\kappa_1/m}$, $\omega_2 = \sqrt{\kappa_2/m}$, and $\omega_3 = \sqrt{\kappa_3/m}$.

In the more general case of finite λ , Hamilton's equations for the P_1, Q_1 conjugate coordinates are

$$\dot{P}_1 = -\frac{\partial H}{\partial Q_1} = -\kappa_1 Q_1 - \lambda Q_2 Q_3, \quad (15.2a)$$

$$\dot{Q}_1 = \frac{\partial H}{\partial P_1} = P_1/m \quad (15.2b)$$

with similar equations for the P_2, Q_2 and P_3, Q_3 conjugates. Using relationships such as $\ddot{Q}_1 = \dot{P}_1/m$, three coupled oscillator equations result, namely

$$\begin{aligned} \ddot{Q}_1 + \omega_1^2 Q_1 &= -\frac{\lambda}{m} Q_2 Q_3 \\ \ddot{Q}_2 + \omega_2^2 Q_2 &= -\frac{\lambda}{m} Q_1 Q_3 \\ \ddot{Q}_3 + \omega_3^2 Q_3 &= -\frac{\lambda}{m} Q_1 Q_2. \end{aligned} \quad (15.3)$$

For small λ , each oscillator may be assumed to oscillate nearly independently so that during each cycle of a given oscillator, the oscillator experiences only slight amplitude and phase changes due to the nonlinear coupling with the other two oscillators. Thus, approximate solutions for the oscillators can be written as

$$\begin{aligned} Q_1(t) &= A_1(t) \cos[\omega_1 t + \delta_1(t)] \\ Q_2(t) &= A_2(t) \cos[\omega_2 t + \delta_2(t)] \\ Q_3(t) &= A_3(t) \cos[\omega_3 t + \delta_3(t)], \end{aligned} \quad (15.4)$$

where it is assumed

$$\frac{\dot{A}_j}{A_j} \ll \omega_j \text{ and } \dot{\delta}_j \ll \omega_j. \quad (15.5)$$

Each oscillator is considered as a linear mode of the system and, from now on, the term mode will be used interchangeably with oscillator. The energy associated with each mode is

$$W_j = \frac{1}{2m} P_j^2 + \frac{1}{2} \kappa_j Q_j^2 = \frac{m}{2} \dot{Q}_j^2 + \frac{1}{2} \kappa_j Q_j^2 = \frac{m}{2} \omega_j^2 A_j^2 \quad (15.6)$$

and the associated action is

$$\begin{aligned} S &= \oint P_j dQ_j = m \oint \dot{Q}_j dQ_j = -m\omega_j A_j^2 \int_0^{2\pi} \sin \psi_j d \cos \psi_j \\ &= m\omega_j A_j^2 \int_0^{2\pi} \sin^2 \psi_j d\psi_j = \frac{m}{2} \omega_j A_j^2, \end{aligned} \quad (15.7)$$

where

$$\psi_j = \omega_j t + \delta_j(t) \quad (15.8)$$

is the phase of the mode.

Let us now use Eq. (15.8) to calculate the first and second derivatives of $Q_j = A_j \cos \psi_j$. Temporarily omitting the j subscript for clarity, the first derivative is seen to be $\dot{Q} = \dot{A} \cos \psi - A \dot{\psi} \sin \psi$ and the second derivative is

$$\begin{aligned} \ddot{Q} &= \ddot{A} \cos \psi - 2\dot{A}\dot{\psi} \sin \psi - A\ddot{\psi} \sin \psi - A\dot{\psi}^2 \cos \psi \\ &= \ddot{A} \cos \psi - 2\dot{A}(\omega + \dot{\delta}) \sin \psi - A\ddot{\delta} \sin \psi - A(\omega + \dot{\delta})^2 \cos \psi. \end{aligned} \quad (15.9)$$

Since the amplitude A and phase δ are assumed to change only slightly during a wave period, ω is very large compared to time derivatives of both the amplitude and the phase, i.e.,

$$\begin{aligned} \frac{d\dot{A}}{dt} &\ll \omega \dot{A} \\ \frac{d\dot{\delta}}{dt} &\ll \omega \dot{\delta}. \end{aligned} \quad (15.10)$$

Thus, any term in Eq. (15.9) involving *two* derivatives of slowly varying quantities will be negligible compared to the corresponding term containing ω and *one* derivative of the slowly varying quantity. In particular, we may drop $\ddot{A}$ compared to $\omega \dot{A}$, $\ddot{\delta}$ compared to $\omega \dot{\delta}$, and $\dot{\delta}^2$ compared to $\omega \dot{\delta}$. On making these WKB-like approximations, Eq. (15.9) becomes

$$\ddot{Q}_j = -\omega_j^2 A_j \cos(\omega_j t + \delta_j) - 2\omega_j \dot{A}_j \sin(\omega_j t + \delta_j) - 2\omega_j A_j \dot{\delta}_j \cos(\omega_j t + \delta_j), \quad (15.11)$$

where the j subscript has now been restored. When the above expression is inserted into Eqs. (15.3), the terms involving ω_j^2 cancel and what remains is

$$\begin{aligned} \left[\begin{array}{l} 2\omega_j \dot{A}_j \sin(\omega_j t + \delta_j) \\ + 2\omega_j A_j \dot{\delta}_j \cos(\omega_j t + \delta_j) \end{array} \right] &= \frac{\lambda}{m} A_k A_l \cos(\omega_k t + \delta_k) \cos(\omega_l t + \delta_l) \\ &= \frac{\lambda}{2m} A_k A_l \left[\begin{array}{l} \cos((\omega_k + \omega_l)t + \delta_k + \delta_l) \\ + \cos((\omega_k - \omega_l)t + \delta_k - \delta_l) \end{array} \right]. \end{aligned} \quad (15.12)$$

It is necessary to write out the three coupled equations explicitly because the coupling terms on the right-hand side are not fully symmetric. To identify resonant interactions it is assumed

$$\omega_3 = \omega_1 + \omega_2 \quad (15.13)$$

$$\theta(t) = \delta_1 + \delta_2 - \delta_3, \quad (15.14)$$

in which case the coupled mode equations can be written

$$\begin{aligned} \begin{bmatrix} \dot{A}_1 \sin(\omega_1 t + \delta_1) \\ + A_1 \dot{\delta}_1 \cos(\omega_1 t + \delta_1) \end{bmatrix} &= \frac{\lambda A_2 A_3}{4m\omega_1} \begin{bmatrix} \cos((\omega_2 + \omega_3)t + \delta_2 + \delta_3) \\ + \underbrace{\cos((\omega_2 - \omega_3)t + \delta_2 - \delta_3)}_{\text{resonant at } -\omega_1} \end{bmatrix} \\ \begin{bmatrix} \dot{A}_2 \sin(\omega_2 t + \delta_2) \\ + A_2 \dot{\delta}_2 \cos(\omega_2 t + \delta_2) \end{bmatrix} &= \frac{\lambda A_1 A_3}{4m\omega_2} \begin{bmatrix} \cos((\omega_1 + \omega_3)t + \delta_1 + \delta_3) \\ + \underbrace{\cos((\omega_1 - \omega_3)t + \delta_1 - \delta_3)}_{\text{resonant at } -\omega_2} \end{bmatrix} \\ \begin{bmatrix} \dot{A}_3 \sin(\omega_3 t + \delta_3) \\ + A_3 \dot{\delta}_3 \cos(\omega_3 t + \delta_3) \end{bmatrix} &= \frac{\lambda A_1 A_2}{4m\omega_3} \begin{bmatrix} \underbrace{\cos((\omega_1 + \omega_2)t + \delta_1 + \delta_2)}_{\text{resonant at } +\omega_3} \\ + \cos((\omega_1 - \omega_2)t + \delta_1 - \delta_2) \end{bmatrix}. \quad (15.15) \end{aligned}$$

Using $\omega_2 - \omega_3 = -\omega_1$, $\delta_2 - \delta_3 = \theta - \delta_1$, $\omega_1 - \omega_3 = -\omega_2$, $\delta_1 - \delta_3 = \theta - \delta_2$, and discarding non-resonant terms, Eq. (15.15) becomes

$$\begin{aligned} \dot{A}_1 \sin(\omega_1 t + \delta_1) + A_1 \dot{\delta}_1 \cos(\omega_1 t + \delta_1) &= \frac{\lambda A_2 A_3}{4m\omega_1} \cos(\theta - \omega_1 t - \delta_1) \\ \dot{A}_2 \sin(\omega_2 t + \delta_2) + A_2 \dot{\delta}_2 \cos(\omega_2 t + \delta_2) &= \frac{\lambda A_1 A_3}{4m\omega_2} \cos(\theta - \omega_2 t - \delta_2) \\ \dot{A}_3 \sin(\omega_3 t + \delta_3) + A_3 \dot{\delta}_3 \cos(\omega_3 t + \delta_3) &= \frac{\lambda A_1 A_2}{4m\omega_3} \cos(\theta + \omega_3 t + \delta_3) \quad (15.16) \end{aligned}$$

or

$$\begin{aligned} \begin{bmatrix} \dot{A}_1 \sin(\omega_1 t + \delta_1) \\ + A_1 \dot{\delta}_1 \cos(\omega_1 t + \delta_1) \end{bmatrix} &= \frac{\lambda A_2 A_3}{4m\omega_1} \begin{bmatrix} \cos \theta \cos(\omega_1 t + \delta_1) \\ + \sin \theta \sin(\omega_1 t + \delta_1) \end{bmatrix} \\ \begin{bmatrix} \dot{A}_2 \sin(\omega_2 t + \delta_2) \\ + A_2 \dot{\delta}_2 \cos(\omega_2 t + \delta_2) \end{bmatrix} &= \frac{\lambda A_1 A_3}{4m\omega_2} \begin{bmatrix} \cos \theta \cos(\omega_2 t + \delta_2) \\ + \sin \theta \sin(\omega_2 t + \delta_2) \end{bmatrix} \\ \begin{bmatrix} \dot{A}_3 \sin(\omega_3 t + \delta_3) \\ + A_3 \dot{\delta}_3 \cos(\omega_3 t + \delta_3) \end{bmatrix} &= \frac{\lambda A_1 A_2}{4m\omega_3} \begin{bmatrix} \cos \theta \cos(\omega_3 t + \delta_3) \\ - \sin \theta \sin(\omega_3 t + \delta_3) \end{bmatrix}. \quad (15.17) \end{aligned}$$

Matching the time-dependent sine and cosine terms on both sides gives

$$\dot{A}_1 = \frac{\lambda A_2 A_3}{4m\omega_1} \sin \theta \quad (15.18a)$$

$$\dot{A}_2 = \frac{\lambda A_1 A_3}{4m\omega_2} \sin \theta \quad (15.18b)$$

$$\dot{A}_3 = -\frac{\lambda A_1 A_2}{4m\omega_3} \sin \theta \quad (15.18c)$$

and

$$\begin{aligned}A_1 \dot{\delta}_1 &= \frac{\lambda A_2 A_3}{4m\omega_1} \cos \theta \\A_2 \dot{\delta}_2 &= \frac{\lambda A_1 A_3}{4m\omega_2} \cos \theta \\A_3 \dot{\delta}_3 &= \frac{\lambda A_1 A_2}{4m\omega_3} \cos \theta.\end{aligned}\tag{15.19}$$

These equations can be used to establish various conservation relations. Multiplying Eqs. (15.18) by $\omega_j^2 A_j$ and summing gives

$$\frac{1}{2} \frac{d}{dt} (\omega_1^2 A_1^2 + \omega_2^2 A_2^2 + \omega_3^2 A_3^2) = \frac{(\omega_1 + \omega_2 - \omega_3) \lambda A_1 A_2 A_3}{4m} \sin \theta = 0$$

so

$$\omega_1^2 A_1^2 + \omega_2^2 A_2^2 + \omega_3^2 A_3^2 = \text{const.}\tag{15.20}$$

This shows that the total energy in the three modes is constant, yet allows portions of this energy to slosh back and forth between modes.

A corresponding set of relations for the action can be obtained by multiplying pairs of equations in Eq. (15.18) by $\omega_j A_j$ and then either adding or subtracting. For example, multiplying the first pair by $\omega_j A_j$ and subtracting gives

$$\frac{1}{2} \frac{d}{dt} (\omega_1 A_1^2 - \omega_2 A_2^2) = 0.\tag{15.21}$$

Appropriate adding and subtracting in this manner gives

$$\begin{aligned}\omega_1 A_1^2 - \omega_2 A_2^2 &= \text{const.} \\ \omega_1 A_1^2 + \omega_3 A_3^2 &= \text{const.} \\ \omega_2 A_2^2 + \omega_3 A_3^2 &= \text{const.}\end{aligned}\tag{15.22}$$

These relationships can be expressed in a manner analogous to quantum principles by defining the effective “quantum number” of a mode as its ratio of energy to frequency,

$$N_j = \frac{W_j}{\omega_j} = \frac{m}{2} \omega_j A_j^2.\tag{15.23}$$

It is clear that N_j is the same as the action except for an unimportant constant factor. Thus, the action conservation rules can be recast as

$$\begin{aligned}N_1 - N_2 &= \text{const.} \\ N_1 + N_3 &= \text{const.} \\ N_2 + N_3 &= \text{const.}\end{aligned}\tag{15.24}$$

or if changes in action are considered

$$\begin{aligned}\Delta N_1 &= +\Delta N_2 \\ \Delta N_1 &= -\Delta N_3 \\ \Delta N_2 &= -\Delta N_3.\end{aligned}\tag{15.25}$$

This provides an action accounting scheme such that a change $\Delta N_3 = -1$ can be considered as a mode 3 “photon” decaying (or equivalently disintegrating) into a mode 1 photon ($\Delta N_1 = +1$) and a mode 2 photon ($\Delta N_2 = +1$), all the while satisfying conservation of energy; this is sketched in Fig. 15.2.

An additional conservation equation can be obtained by subtracting the last of Eqs. (15.19) from the sum of the first two to obtain

$$\begin{aligned}\dot{\theta} &= \dot{\delta}_1 + \dot{\delta}_2 - \dot{\delta}_3 \\ &= \left(\frac{\lambda A_2 A_3}{4m A_1 \omega_1} + \frac{\lambda A_1 A_3}{4m A_2 \omega_2} - \frac{\lambda A_1 A_2}{4m A_3 \omega_3}\right) \cos \theta \\ &= \left(\frac{\dot{A}_1}{A_1} + \frac{\dot{A}_2}{A_2} + \frac{\dot{A}_3}{A_3}\right) \frac{\cos \theta}{\sin \theta}\end{aligned}\tag{15.26}$$

and then integrating to find

$$A_1 A_2 A_3 \cos \theta = const.\tag{15.27}$$

We now consider some solutions to the system of equations given by Eqs. (15.18a)–(15.18c). Suppose that initially $A_3 \gg A_2, A_1$. In this case Eq.(15.18c) gives $A_3 \simeq const.$ Solving Eq. (15.18b) for A_1 and substituting the

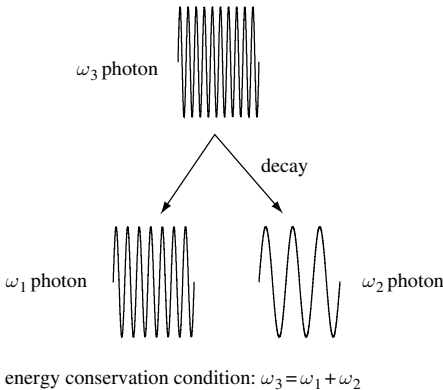


Fig. 15.2 Photon at frequency ω_3 decaying into a photon at frequency ω_1 and a photon at frequency ω_2 .

result in Eq. (15.18a) gives

$$\frac{1}{\sin \theta} \frac{d}{dt} \left(\frac{1}{\sin \theta} \frac{dA_2}{dt} \right) = \frac{\lambda^2 A_3^2}{16m^2 \omega_1 \omega_2} A_2, \quad (15.28)$$

which has exponentially growing solutions if $\omega_1 \omega_2 > 0$. By defining

$$\tau = \int_0^t dt' \sin \theta(t') \quad (15.29)$$

it is seen that

$$\frac{dA_2}{dt} = \frac{d\tau}{dt} \frac{dA_2}{d\tau} = \sin \theta \frac{dA_2}{d\tau} \quad (15.30)$$

and so

$$\frac{d^2 A_2}{d\tau^2} = \frac{\lambda^2 A_3^2}{16m^2 \omega_1 \omega_2} A_2. \quad (15.31)$$

Since $\omega_1 \omega_2 > 0$ and $\omega_1 + \omega_2 = \omega_3$, if ω_3 is chosen to be positive then both ω_1 and ω_2 must both be positive and so $\omega_3 > \omega_1, \omega_2$. If A_2 grows, then, because of the action rules, A_1 must grow in the same proportion. Furthermore, as A_1 and A_2 grow, A_3 must decrease until the approximation $A_3 \gg A_1, A_2$ fails. This process can be considered as a high-energy “photon” with frequency ω_3 decaying into an ω_1 photon and an ω_2 photon where the last two photons have lower energy than ω_1 . Equation (15.31) shows that the decay of the mode 3 photon into mode 1,2 photons can be characterized by an exponential growth of the mode 2 amplitude,

$$A_2 \sim e^{\gamma \tau}, \quad (15.32)$$

where

$$\gamma = \frac{\lambda A_3}{4m \sqrt{\omega_1 \omega_2}} \quad (15.33)$$

is the instability growth rate. Mode 3 is called the pump mode since it supplies the energy for the instability. Modes 1 and 2 are called the daughter modes.

15.3 Application to waves

Nonlinearities in wave equations give rise to coupled systems of equations similar to Eqs. (15.3). Suppose Q now refers to a plasma parameter, say density, and that the plasma can support three distinct waves (modes) having respective linear density fluctuations

$$\begin{aligned} Q_1(\mathbf{x}, t) &= A_1(t) \cos(\mathbf{k}_1 \cdot \mathbf{x} - \omega_1 t - \delta_1(t)) \\ Q_2(\mathbf{x}, t) &= A_2(t) \cos(\mathbf{k}_2 \cdot \mathbf{x} - \omega_2 t - \delta_2(t)) \\ Q_3(\mathbf{x}, t) &= A_3(t) \cos(\mathbf{k}_3 \cdot \mathbf{x} - \omega_3 t - \delta_3(t)); \end{aligned} \quad (15.34)$$

the density Q therefore has the functional form

$$Q(\mathbf{x}, t) = Q_1(\mathbf{x}, t) + Q_2(\mathbf{x}, t) + Q_3(\mathbf{x}, t). \quad (15.35)$$

This situation is mathematically analogous to the coupled oscillator system discussed in Section 15.2 if the amplitude Q_j of each of the three waves is considered to be an effective canonical coordinate. Each wave satisfies a linear dispersion relation $\omega_j = \omega_j(\mathbf{k})$ and nonlinearities in the wave equations provide a mutual coupling between the modes analogous to the coupling between the different directions of motion for the oscillating mass discussed in Section 15.2. The $\exp(i\mathbf{k} \cdot \mathbf{x})$ dependence of the waves suggests the need for a wavenumber selection rule

$$\mathbf{k}_3 = \mathbf{k}_1 + \mathbf{k}_2 \quad (15.36)$$

analogous to the frequency selection rule Eq. (15.13). Using the wavenumber selection rule and generalizing the phase offset definition to be $\delta'_j(t) = \delta_j(t) - \mathbf{k}_j \cdot \mathbf{x}$, it is seen that the coupled wave equations become identical to the coupled oscillator equations with δ'_j replacing δ_j . Hence, so long as the selection rules $\omega_3 = \omega_1 + \omega_2$ and $\mathbf{k}_3 = \mathbf{k}_1 + \mathbf{k}_2$ are satisfied, a high-frequency wave ω_3 should decay spontaneously into two low-frequency waves ω_1, ω_2 , providing a suitable coupling coefficient exists.

15.3.1 Examples of nonlinearities

Nonlinearities can be divided into two general types: (1) two high-frequency waves beating together to give a low-frequency driving term and (2) a high-frequency wave beating with a low-frequency wave to give a high-frequency driving term. As shown below, the first type of nonlinearity has the form of a radiation pressure (also known as a ponderomotive force) while the second type of nonlinearity has the form of a density modulation.

1. Beat of two high-frequency waves driving a low-frequency wave (ponderomotive force).

Since ion motion is negligible in a high-frequency wave, all that is required when considering nonlinearities is the electron equation of motion,

$$\frac{\partial \mathbf{u}_e}{\partial t} = -\mathbf{u}_e \cdot \nabla \mathbf{u}_e + \frac{q_e}{m_e} (\mathbf{E} + \mathbf{u}_e \times \mathbf{B}) - \frac{1}{m_e n_e} \nabla P_e. \quad (15.37)$$

For simplicity, it is assumed that no equilibrium magnetic field exists, so the only magnetic field is the wave magnetic field. Tildes are used to denote linear quantities to avoid confusion with the subscripts 1, 2, 3, which denote the low-frequency daughter, high-frequency daughter, and pump wave respectively. If the high-frequency wave is electromagnetic, then the mode is transverse, i.e., the wavevector $\mathbf{k}$ is perpendicular to the mode electric field $\tilde{\mathbf{E}}$ and so there is no coupling to pressure oscillations because

$\nabla \tilde{P}_e$ is in the direction of $\mathbf{k}$. If the high-frequency mode is electrostatic, i.e., is a Langmuir wave, then there is a finite $\nabla \tilde{P}_e$ contribution, but this term is small because of the adiabatic assumption $\omega/k \gg v_{Te}$ for a Langmuir wave (see Eq. (4.28)) and so can be ignored to first approximation. Thus, when calculating the linear motion for purposes of determining nonlinear coupling coefficients, the $\nabla \tilde{P}_e$ term resulting from linearizing Eq. (15.37) may be dropped and so we simply have

$$\frac{\partial \tilde{\mathbf{u}}_e}{\partial t} = \frac{q_e}{m_e} \tilde{\mathbf{E}}. \quad (15.38)$$

This is exactly true for an electromagnetic wave and true to lowest order in the adiabatic assumption for an electrostatic wave. The electron quiver velocity is defined to be

$$\tilde{\mathbf{u}}_e^h = \frac{q_e}{m_e} \int^t \tilde{\mathbf{E}} dt', \quad (15.39)$$

which is the solution to Eq. (15.38) for both electron plasma waves and electromagnetic waves. The main nonlinear terms in Eq. (15.37) are $-\tilde{\mathbf{u}}_e \cdot \nabla \tilde{\mathbf{u}}_e + q_e (\tilde{\mathbf{u}}_e \times \tilde{\mathbf{B}})/m_e$; nonlinearity in the pressure gradient is ignored because, by assumption, this term is already small. In order to obtain $\tilde{\mathbf{B}}$, Faraday's law is integrated with respect to time giving

$$\tilde{\mathbf{B}} = -\nabla \times \int^t \tilde{\mathbf{E}} dt' = -\frac{m_e}{q_e} \nabla \times \tilde{\mathbf{u}}_e^h. \quad (15.40)$$

The two nonlinear terms can be combined using Eq. (15.40) to form what is called the ponderomotive force or radiation pressure,

$$\begin{aligned} -\tilde{\mathbf{u}}_e \cdot \nabla \tilde{\mathbf{u}}_e + \frac{q_e}{m_e} (\tilde{\mathbf{u}}_e \times \tilde{\mathbf{B}}) &= -\tilde{\mathbf{u}}_e^h \cdot \nabla \tilde{\mathbf{u}}_e^h - (\tilde{\mathbf{u}}_e^h \times \nabla \times \tilde{\mathbf{u}}_e^h) \\ &= -\frac{1}{2} \nabla (\tilde{u}_e^h)^2. \end{aligned} \quad (15.41)$$

If only one high-frequency mode exists and beats with itself, then the ponderomotive force $-\nabla (\tilde{u}_e^h)^2/2$ is at zero frequency but if the beating is between two distinct high-frequency modes, then the ponderomotive force contains a term at the difference (i.e., beat) frequency between the frequencies of the two modes. Thus, at low frequencies the electron equation of motion, Eq. (15.37) becomes

$$\frac{\partial \tilde{\mathbf{u}}_e}{\partial t} = \frac{q_e}{m_e} \tilde{\mathbf{E}} - \frac{1}{m_e n} \nabla \tilde{P}_e - \frac{1}{2} \nabla (\tilde{u}_e^h)^2, \quad (15.42)$$

where only the beat frequency component in $(\tilde{u}_e^h)^2$ is used. The ponderomotive force provides a mechanism for high-frequency waves to couple to low-frequency waves. It acts as an effective pressure scaling as $m_e n (\tilde{u}_e^h)^2/2$ and so, in a sense, the quiver velocity $\tilde{u}_e^h$ acts as a thermal velocity. The ratio of ion radiation pressure to the ion pressure is smaller than the corresponding ratio for electrons by a factor of m_e/m_i because the ion quiver velocity is smaller by this factor. Thus, ion ponderomotive force is ignored since it is so small.

2. *Beating of a low-frequency wave with a high-frequency wave to drive another high-frequency wave (modulation).* By writing Ampère's law as

$$\nabla \times \tilde{\mathbf{B}} = \mu_0 \sum_{\sigma} (\tilde{n}_{\sigma} q_{\sigma} \tilde{\mathbf{u}}_{\sigma} + n_0 q_{\sigma} \tilde{\mathbf{u}}_{\sigma}) + \mu_0 \varepsilon_0 \frac{\partial \tilde{\mathbf{E}}}{\partial t} \quad (15.43)$$

it is seen that density fluctuations provide a nonlinear component to the current density. The nonlinear term can be put on the right-hand side to emphasize its role as a nonlinear driving term so that Ampère's law becomes

$$\nabla \times \tilde{\mathbf{B}} - \mu_0 \varepsilon_0 \frac{\partial \tilde{\mathbf{E}}}{\partial t} - \mu_0 \sum_{\sigma} n_0 q_{\sigma} \tilde{\mathbf{u}}_{\sigma} = -\mu_0 \sum_{\sigma} \tilde{n}_{\sigma} q_{\sigma} \tilde{\mathbf{u}}_{\sigma}. \quad (15.44)$$

The nonlinear term is assumed to be a product of a high-frequency wave and a low-frequency wave. The linearized continuity equation gives

$$\frac{\partial \tilde{n}_{\sigma}}{\partial t} = -n_{\sigma} \nabla \cdot \tilde{\mathbf{u}}_{\sigma} \quad (15.45)$$

showing that $\tilde{n}_{\sigma}/n_{\sigma} = \mathbf{k} \cdot \tilde{\mathbf{u}}_{\sigma}/\omega$. The low-frequency wave typically has a smaller phase velocity than the high-frequency wave so, in terms of magnitudes,

$$\frac{\tilde{n}_{\sigma}^l}{\tilde{n}_{\sigma}^h} \sim \frac{\tilde{u}_{\sigma}^l (\omega/k)_h}{\tilde{u}_{\sigma}^h (\omega/k)_l} \gg \frac{\tilde{u}_{\sigma}^l}{\tilde{u}_{\sigma}^h}. \quad (15.46)$$

Thus, the product $\tilde{n}_{\sigma}^l \tilde{\mathbf{u}}_{\sigma}^h$ is much larger than the product $\tilde{n}_{\sigma}^h \tilde{\mathbf{u}}_{\sigma}^l$, where l and h refer to low- and high-frequency waves. Thus, the dominant effect of a low-frequency wave is to modulate the density profile seen by a high-frequency wave.

15.3.2 Possible types of wave interaction

As discussed in Section 4.2, three distinct types of waves can propagate in an unmagnetized uniform plasma and these waves have the dispersion relations:

$$\begin{aligned} \omega^2 &= \omega_{pe}^2 + k^2 c^2, \text{ electromagnetic wave} \\ \omega^2 &= \omega_{pe}^2 (1 + 3k^2 \lambda_{de}^2), \text{ electron plasma wave} \\ \omega^2 &= \frac{k^2 c_s^2}{1 + k^2 \lambda_{De}^2}, \text{ ion acoustic wave.} \end{aligned} \quad (15.47)$$

If three waves satisfy the selection rules

$$\omega_3 = \omega_1 + \omega_2 \quad (15.48)$$

$$\mathbf{k}_3 = \mathbf{k}_1 + \mathbf{k}_2, \quad (15.49)$$

they can only have the *same* dispersion relation if ω^2 depends linearly on k^2 so that the magnitude of $\mathbf{k}$ is linearly proportional to the magnitude of ω . This is not so for the modes listed in Eq. (15.47) except for the $k^2 \lambda_{De}^2 \ll 1$ limit of the ion

Table 15.1

Pump	Hf daughter	Lf daughter	Common name
ω_3	ω_2	ω_1	
em	em	Langmuir	stimulated Raman backscatter
em	em	acoustic	stimulated Brillouin backscatter
em	em	zero frequency	self-focusing
em	Langmuir	Langmuir	two-plasmon decay
em	Langmuir	acoustic	parametric decay instability
Langmuir	Langmuir	acoustic	electron decay instability
Langmuir	Langmuir	zero frequency	caviton

acoustic wave; applying the selection rules to this $\omega^2 = k^2 c_s^2$ limit of ion acoustic waves gives nonlinear interactions between various sound wave harmonics, e.g., the first and second harmonic can interact with the third harmonic. Since this limit corresponds to the well-known nonlinear steepening of ordinary sound waves it will not be discussed further.

The situations that are of particular relevance to plasmas are where modes satisfying different types of dispersion relations interact with each other, for example an electromagnetic wave interacting with an electron plasma wave and an ion acoustic wave. The various possibilities for nonlinear interactions of the three types of plasma waves are shown in Table 15.1.

Here the abbreviations em, Langmuir, and acoustic stand for electromagnetic wave, electron plasma wave, and ion acoustic wave respectively and the ordering is by progressively lower frequency, column by column, taking into account (i) $\omega_3 > \omega_2 > \omega_1$ and (ii) electromagnetic waves have higher frequencies than Langmuir waves, which in turn have higher frequencies than ion acoustic waves.

In each case the low-frequency daughter modulates the density and beats with either the pump or the high-frequency daughter to provide a high-frequency nonlinear current as given by Eq. (15.44) while the pump and the high-frequency daughter beat together to provide a ponderomotive force that couples to the low-frequency wave. Consider a coordinate system where the x axis is in the direction of $\mathbf{k}_3$ and $\perp$ refers to the direction perpendicular to $\mathbf{k}_3$. Thus, $(\mathbf{k}_1 + \mathbf{k}_2)_\perp = 0$ so any daughter wavevector components in the direction perpendicular to the pump wave must be equal in magnitude and opposite in direction.

The high-frequency wave has a faster phase velocity than the daughter waves so

$$\frac{\omega_3^2}{k_3^3} > \frac{\omega_2^2}{k_2^2} \text{ and } \frac{\omega_3^2}{k_3^3} > \frac{\omega_1^2}{k_1^2}. \quad (15.50)$$

Let us suppose that all waves are in the same direction so we may assume k_1, k_2 and k_3 , are all positive. Then taking the square root of all quantities in Eq. (15.50) and using $\omega_3 = \omega_1 + \omega_2$, $k_3 = k_1 + k_2$ gives

$$\frac{\omega_1 + \omega_2}{k_1 + k_2} > \frac{\omega_2}{k_2} \text{ and } \frac{\omega_1 + \omega_2}{k_1 + k_2} > \frac{\omega_1}{k_1}. \quad (15.51)$$

However, rearranging these two expressions leads to the contradictory relationships

$$\frac{\omega_1}{\omega_2} > \frac{k_1}{k_2} \text{ and } \frac{\omega_1}{\omega_2} < \frac{k_1}{k_2}. \quad (15.52)$$

This means it is impossible to have all waves in the same direction and satisfy the selection rules when the pump wave is faster than the daughter waves. Thus, the frequency and wavenumber selection rules can only be satisfied if $\mathbf{k}_1$, $\mathbf{k}_2$, and $\mathbf{k}_3$ are not all in the same direction.

If the daughter waves have no wavevector components perpendicular to the pump wave, then one of the daughter waves must have a wavevector antiparallel to the pump while the other is parallel to the pump. This is shown schematically in Fig. 15.3 for the parametric decay instability where it is seen that the ion acoustic daughter wave propagates backwards relative to the em and Langmuir waves. Since $v_{Te} \ll c$, the phase velocity of the electromagnetic wave is actually much faster than suggested by the figure, i.e., the speed of light is shown artificially slowed down in the figure to make the vector relationships more obvious. In actual parametric decay situations the approximation $k_1 + k_2 \simeq 0$ may be used since $|k_3| \ll |k_1|, |k_2|$.

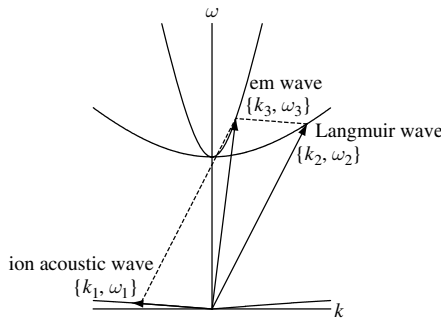


Fig. 15.3 Electromagnetic wave $\omega = \sqrt{\omega_{pe}^2 + k^2 c^2}$ decaying into a Langmuir wave $\omega = \omega_{pe} \sqrt{1 + 3k^2 \lambda_{De}^2}$ and an ion acoustic wave $\omega = kc_s / \sqrt{1 + k^2 \lambda_{De}^2}$. The ion acoustic wave propagates backwards in order to have $\omega_3(k_1 + k_2) = \omega_1(k_1) + \omega_2(k_2)$ as can be seen from the addition of the $\{k, \omega\}$ vectors in the diagram.

The electromagnetic wave in a given plasma always has a higher frequency than the electron plasma or ion acoustic waves. As a consequence, the low-frequency daughter cannot be an electromagnetic wave because then all three waves would have to be electromagnetic waves, which would then violate the requirement that the dispersion relations cannot all be the same. The low-frequency daughter must therefore be either an ion acoustic wave or an electron plasma wave. Similarly, the high-frequency wave (pump or high-frequency daughter) cannot be an ion acoustic wave and so must be either an electromagnetic wave or an electron plasma wave. Thus, the various interactions tabulated above can be accounted for by establishing the appropriate coupling equations for the following four possibilities: high-frequency wave is an electromagnetic wave or a Langmuir wave, low-frequency wave is a Langmuir wave or an ion acoustic wave.

Low-frequency wave is an ion acoustic wave

On assuming quasi-neutrality, which corresponds to assuming $k^2 \lambda_{De}^2 \ll 1$, the low frequency electron and ion equations of motion may be approximated as

$$\frac{\partial \tilde{\mathbf{u}}_e}{\partial t} = \frac{q_e}{m_e} \tilde{\mathbf{E}} - \frac{\kappa T_e}{m_e n} \nabla \tilde{n} - \frac{1}{2} \nabla (\tilde{u}_e^h)^2 \quad (15.53)$$

$$\frac{\partial \tilde{\mathbf{u}}_i}{\partial t} = \frac{q_i}{m_i} \tilde{\mathbf{E}}; \quad (15.54)$$

here only the low-frequency beat component of $(\tilde{u}_e^h)^2$ is used. Also, as shown earlier, the ion ponderomotive force is negligible and therefore ignored. Because the electron mass is very small, the left-hand side of the electron equation of motion is dropped, in which case this equation reduces to the simple force balance relation

$$\frac{q_e}{m_e} \tilde{\mathbf{E}} - \frac{\kappa T_e}{m_e n} \nabla \tilde{n} - \frac{1}{2} \nabla (\tilde{u}_e^h)^2 \simeq 0. \quad (15.55)$$

Using Eq. (15.55) to eliminate $\tilde{\mathbf{E}}$ from the ion equation gives

$$\frac{\partial \tilde{\mathbf{u}}_i}{\partial t} = -\frac{\kappa T_e}{m_i n} \nabla \tilde{n} - \frac{1}{2} \frac{m_e}{m_i} \nabla (\tilde{u}_e^h)^2. \quad (15.56)$$

Because quasi-neutrality is assumed, $\tilde{n}_i = \tilde{n}_e = \tilde{n}$, and so the time derivative of the ion continuity equation can be written as

$$\frac{\partial^2 \tilde{n}}{\partial t^2} + n \nabla \cdot \frac{\partial \tilde{\mathbf{u}}_i}{\partial t} = 0. \quad (15.57)$$

On substituting for $\partial \tilde{\mathbf{u}}_i / \partial t$, the above equation becomes

$$\frac{\partial^2 \tilde{n}}{\partial t^2} - n \nabla \cdot \left(\frac{\kappa T_e}{m_i n} \nabla \tilde{n} + \frac{1}{2} \frac{m_e}{m_i} \nabla (\tilde{u}_e^h)^2 \right) = 0, \quad (15.58)$$

which can be written as an ion acoustic wave equation with a nonlinear coupling term due to the electron ponderomotive force,

$$\frac{\partial^2 \tilde{n}}{\partial t^2} - c_s^2 \nabla^2 \tilde{n} = \frac{n}{2} \frac{m_e}{m_i} \nabla^2 (\tilde{u}_e^h)^2. \quad (15.59)$$

If the electron quiver velocity is considered to behave as an effective thermal velocity, then the low-frequency beat component of $(\tilde{u}_e^h)^2$ can be considered as modulating the effective electron temperature. Specifically,

$$\tilde{n}_e \kappa T_e + n \kappa \tilde{T}_e = \tilde{n}_e \kappa T_e + n \frac{m_e (\tilde{u}_e^h)^2}{2} \quad (15.60)$$

and so

$$\tilde{n} c_s^2 \rightarrow \tilde{n} c_s^2 + n (\widetilde{c_s^2}) = \tilde{n} c_s^2 + n \frac{\kappa \tilde{T}_e}{m_i} = \tilde{n} c_s^2 + n \frac{m_e (\tilde{u}_e^h)^2}{2 m_i}, \quad (15.61)$$

which is consistent with Eq. (15.59).

Low-frequency wave is an electron plasma wave

In this case the low-frequency wave frequency is above ω_{pe} so the ions cannot respond and therefore can be considered to be a stationary, neutralizing background. Now only the electron dynamic response matters and the density perturbation is not quasi-neutral. The electron equation of motion for the low-frequency wave is

$$\frac{\partial \tilde{\mathbf{u}}_e}{\partial t} = \frac{q_e}{m_e} \tilde{\mathbf{E}} - \frac{3\kappa T_e}{m_e n} \nabla \tilde{n}_e - \frac{1}{2} \nabla (\tilde{u}_e^h)^2, \quad (15.62)$$

where the 3 comes from the adiabatic pressure perturbation. Combining this with the time derivative of the electron continuity equation,

$$\frac{\partial^2 \tilde{n}_e}{\partial t^2} + n \nabla \cdot \left(\frac{\partial \tilde{\mathbf{u}}_e}{\partial t} \right) = 0, \quad (15.63)$$

gives

$$\frac{\partial^2 \tilde{n}_e}{\partial t^2} + n \nabla \cdot \left(\frac{q_e}{m_e} \tilde{\mathbf{E}} - \frac{3\kappa T_e}{m_e n} \nabla \tilde{n}_e - \frac{1}{2} \nabla (\tilde{u}_e^h)^2 \right) = 0. \quad (15.64)$$

This may be simplified by invoking Poisson's equation

$$\nabla \cdot \tilde{\mathbf{E}} = \frac{1}{\epsilon_0} \tilde{n}_e q_e \quad (15.65)$$

to obtain

$$\frac{\partial^2 \tilde{n}_e}{\partial t^2} + \omega_{pe}^2 \tilde{n}_e - \frac{3\kappa T_e}{m_e} \nabla^2 \tilde{n}_e = \frac{n}{2} \nabla^2 (\tilde{u}_e^h)^2. \quad (15.66)$$

The left-hand side is the electron plasma wave equation (Langmuir wave) and the right-hand side provides the nonlinear drive (or coupling) due to ponderomotive force. Again, using Eq. (15.60) it is seen that the ponderomotive force term acts like a modulation of the electron temperature such that $\widehat{n\kappa T_e} \rightarrow \tilde{n}\kappa T_e + n\kappa \tilde{T}_e$, where $\tilde{T}_e$ is due to the high-frequency electron quiver velocity.

High-frequency wave is an electron plasma wave

In this case the wave is electrostatic and so there is no high-frequency oscillating magnetic field. The nonlinear continuity equation is

$$\frac{\partial \tilde{n}_e}{\partial t} + n \nabla \cdot \tilde{\mathbf{u}}_e = -\nabla \cdot (\tilde{n}_e \tilde{\mathbf{u}}_e) \quad (15.67)$$

and taking a time derivative this becomes

$$\frac{\partial^2 \tilde{n}_e}{\partial t^2} + n \nabla \cdot \frac{\partial \tilde{\mathbf{u}}_e}{\partial t} = -\frac{\partial}{\partial t} \nabla \cdot (\tilde{n}_e \tilde{\mathbf{u}}_e). \quad (15.68)$$

However, the high-frequency linear electron equation of motion is

$$\frac{\partial \tilde{\mathbf{u}}_e}{\partial t} = \frac{q_e}{m_e} \tilde{\mathbf{E}} - \frac{3\kappa T_e}{m_e n} \nabla \tilde{n}_e \quad (15.69)$$

so Eq. (15.68) becomes

$$\frac{\partial^2 \tilde{n}_e}{\partial t^2} + \omega_{pe}^2 \tilde{n}_e - \frac{3\kappa T_e}{m_e} \nabla^2 \tilde{n}_e = -\frac{\partial}{\partial t} \nabla \cdot (\tilde{n}_e \tilde{\mathbf{u}}_e). \quad (15.70)$$

In principle, there are two possible components in the right-hand side nonlinear term since $\tilde{n}_e \tilde{\mathbf{u}}_e = \tilde{n}_e^l \tilde{\mathbf{u}}_e^h + \tilde{n}_e^h \tilde{\mathbf{u}}_e^l$, where l and h refer to low- and high-frequency waves. However, as was shown by Eq. (15.46), the $\tilde{n}_e^l \tilde{\mathbf{u}}_e^h$ term is the dominant term and so the $\tilde{n}_e^h \tilde{\mathbf{u}}_e^l$ term can be discarded. If the high-frequency wave associated with $\tilde{\mathbf{u}}_e^h$ is an electromagnetic wave, then the electric field is transverse so $\nabla \cdot \tilde{\mathbf{u}}_e^h \sim \nabla \cdot \tilde{\mathbf{E}}^h = 0$ and, in this case, $\tilde{n}_e^h \tilde{\mathbf{u}}_e^l$ is not just small, but in fact is exactly zero since $\tilde{n}_e^h = 0$. Hence, the high-frequency wave equation becomes

$$\frac{\partial^2 \tilde{n}_e}{\partial t^2} + \omega_{pe}^2 \tilde{n}_e - \frac{3\kappa T_e}{m_e} \nabla^2 \tilde{n}_e = -\frac{\partial}{\partial t} \nabla \cdot (\tilde{n}_e^l \tilde{\mathbf{u}}_e^h). \quad (15.71)$$

If the high-frequency wave is electromagnetic, then $\nabla \cdot (\tilde{n}_e^l \tilde{\mathbf{u}}_e^h) = \tilde{\mathbf{u}}_e^h \cdot \nabla \tilde{n}_e^l$ showing that the nonlinearity consists of a density modulation due to the high-frequency

motion across the density ripples of the low-frequency mode. For purposes of consistency, it is worthwhile to express Eq. (15.71) in terms of electric field using

$$\nabla \cdot \tilde{\mathbf{E}} = \frac{1}{\epsilon_0} \tilde{n}_e q_e \quad (15.72)$$

so

$$\nabla \cdot \left(\frac{\partial^2 \tilde{\mathbf{E}}}{\partial t^2} + \omega_{pe}^2 \tilde{\mathbf{E}} - \frac{3\kappa T_e}{m_e} \nabla^2 \tilde{\mathbf{E}} \right) = -\frac{\partial}{\partial t} \frac{1}{\epsilon_0} \nabla \cdot (\tilde{n}_e^l q_e \tilde{\mathbf{u}}_e^h), \quad (15.73)$$

which can be integrated in space to give the general expression for a high-frequency electron plasma wave with nonlinear coupling term,

$$\frac{\partial^2 \tilde{\mathbf{E}}}{\partial t^2} + \omega_{pe}^2 \tilde{\mathbf{E}} - \frac{3\kappa T_e}{m_e} \nabla^2 \tilde{\mathbf{E}} = -\frac{1}{\epsilon_0} \frac{\partial}{\partial t} (\tilde{n}_e^l q_e \tilde{\mathbf{u}}_e^h). \quad (15.74)$$

High-frequency wave is an electromagnetic wave

Here the wave is transverse so $\nabla \cdot \tilde{\mathbf{E}} = 0$ and the curl of Faraday's law becomes

$$\nabla^2 \tilde{\mathbf{E}} = \frac{\partial \nabla \times \tilde{\mathbf{B}}}{\partial t}. \quad (15.75)$$

Substituting Eq. (15.44) and using the linear equation of motion gives the general expression for a high-frequency electromagnetic wave with nonlinear coupling term,

$$\frac{\partial^2 \tilde{\mathbf{E}}}{\partial t^2} + \omega_{pe}^2 \tilde{\mathbf{E}} - c^2 \nabla^2 \tilde{\mathbf{E}} = -\frac{1}{\epsilon_0} \frac{\partial}{\partial t} (\tilde{n}_e^l q_e \tilde{\mathbf{u}}_e^h); \quad (15.76)$$

the term $\tilde{n}_e^h q_e \tilde{\mathbf{u}}_e^l$ has been dropped for the reasons given in the previous paragraph.

Summary of mode interactions

Examination of the various combinations considered above shows that the nonlinear coupling acting on high-frequency modes (either pump or high-frequency daughter) is an effective modulation of the density experienced by the high-frequency wave, i.e., $\omega_{pe}^2 \rightarrow \omega_{pe}^2 + \omega_{pe}^2 \tilde{n}^l/n$. On the other hand, the nonlinear coupling acting on the low-frequency daughter comes from the electron ponderomotive force, which effectively modulates the electron temperature experienced by the low-frequency mode, i.e., $\tilde{n}\kappa T_e \rightarrow \tilde{n}\kappa T_e + n\kappa m (\tilde{u}_e^h)^2/2$.

Coupled oscillator formulation

To see how the coupled wave equations can be expressed in terms of coupled oscillators, the specific situation of an electromagnetic wave interacting with a

Langmuir wave and an ion acoustic wave is now considered. In this case, the three coupled equations are

$$\begin{aligned}\frac{\partial^2 \tilde{\mathbf{E}}_3}{\partial t^2} + \omega_{pe}^2 \tilde{\mathbf{E}}_3 - c^2 \nabla^2 \tilde{\mathbf{E}}_3 &= -\frac{q_e}{\varepsilon_0} \frac{\partial}{\partial t} (\tilde{n}_1 \tilde{\mathbf{u}}_2) \\ \frac{\partial^2 \tilde{\mathbf{E}}_2}{\partial t^2} + \omega_{pe}^2 \tilde{\mathbf{E}}_2 - \frac{3\kappa T_e}{m_e} \nabla^2 \tilde{\mathbf{E}}_2 &= -\frac{q_e}{\varepsilon_0} \frac{\partial}{\partial t} (\tilde{n}_1 \tilde{\mathbf{u}}_3) \\ \frac{\partial^2 \tilde{n}_1}{\partial t^2} - c_s^2 \nabla^2 \tilde{n}_1 &= n \frac{m_e}{m_i} \nabla^2 \langle \tilde{\mathbf{u}}_2 \cdot \tilde{\mathbf{u}}_3 \rangle,\end{aligned}\quad (15.77)$$

where the relation

$$\left(\frac{1}{2} (\tilde{u}_e^h)^2 \right)_{\omega_1, \mathbf{k}_1} = \frac{1}{2} \langle (\tilde{\mathbf{u}}_2 + \tilde{\mathbf{u}}_3) \cdot (\tilde{\mathbf{u}}_2 + \tilde{\mathbf{u}}_3) \rangle = \langle \tilde{\mathbf{u}}_2 \cdot \tilde{\mathbf{u}}_3 \rangle \quad (15.78)$$

has been used and the angle brackets refer to the component oscillating at the beat frequency $\omega_1 = \omega_3 - \omega_2$ and having the beat wavevector $\mathbf{k}_1 = \mathbf{k}_3 - \mathbf{k}_2$.

The subscript e has been dropped from all dependent variables because they all refer to electrons. Since the acoustic wave frequency is much smaller than ω_{pe} , it is possible to approximate

$$\frac{\partial}{\partial t} (\tilde{n}_1 \tilde{\mathbf{u}}_2) \simeq \tilde{n}_1 \frac{\partial \tilde{\mathbf{u}}_2}{\partial t} = \tilde{n}_1 \frac{q_e}{m_e} \tilde{\mathbf{E}}_2. \quad (15.79)$$

Using the quiver relation Eq. (15.39), the product of the high-frequency velocities can be expressed as

$$\langle \tilde{\mathbf{u}}_2 \cdot \tilde{\mathbf{u}}_3 \rangle = \frac{q_e^2}{m_e^2 \omega_2 \omega_3} \langle \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3 \rangle. \quad (15.80)$$

Although the two velocities are oscillating 90° out of phase relative to their respective electric fields, the nonlinear product of the velocities has the same phase behavior as the corresponding electric fields. This is because the nonlinear product of the velocities scales as $\langle \cos \omega_2 t \cos \omega_3 t \rangle$ and the nonlinear product of the electric fields scales as $\langle \sin \omega_2 t \sin \omega_3 t \rangle$ so that the difference between these scalings is $\langle \cos \omega_2 t \cos \omega_3 t - \sin \omega_2 t \sin \omega_3 t \rangle = \langle \cos(\omega_2 + \omega_3)t \rangle$, which vanishes because $\omega_2 + \omega_3$ is non-resonant. The system of equations thus becomes

$$\begin{aligned}\left(\frac{\partial^2}{\partial t^2} + \omega_{pe}^2 - c^2 \nabla^2 \right) \tilde{\mathbf{E}}_3 &= -\omega_{pe}^2 \frac{\tilde{n}_1}{n} \tilde{\mathbf{E}}_2 \\ \left(\frac{\partial^2}{\partial t^2} + \omega_{pe}^2 - \frac{3\kappa T_e}{m_e} \nabla^2 \right) \tilde{\mathbf{E}}_2 &= -\omega_{pe}^2 \frac{\tilde{n}_1}{n} \tilde{\mathbf{E}}_3 \\ \left(\frac{\partial^2}{\partial t^2} - c_s^2 \nabla^2 \right) \frac{\tilde{n}_1}{n} &= \frac{q_e^2}{m_i m_e \omega_2 \omega_3} \nabla^2 \langle \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3 \rangle.\end{aligned}\quad (15.81)$$

If different modes are used, e.g., a Langmuir mode decaying into another Langmuir mode, or if the low-frequency mode is a Langmuir wave, the left-hand side wave terms will be changed accordingly, but the right-hand coupling terms will stay the same except if the low-frequency wave is a Langmuir wave, in which case the product $m_i m_e$ must be replaced by m_e^2 in the denominator of the right-hand side of the third equation.

The right-hand coupling terms can be made identical by defining a renormalized density perturbation

$$\tilde{\psi} = \alpha \frac{\tilde{n}_1}{n} \quad (15.82)$$

so the equations become

$$\begin{aligned} \left(\frac{\partial^2}{\partial t^2} + \omega_{pe}^2 - c^2 \nabla^2 \right) \tilde{\mathbf{E}}_3 &= -\omega_{pe}^2 \frac{\tilde{\psi}}{\alpha} \tilde{\mathbf{E}}_2 \\ \left(\frac{\partial^2}{\partial t^2} + \omega_{pe}^2 - \frac{3\kappa T_e}{m_e} \nabla^2 \right) \tilde{\mathbf{E}}_2 &= -\omega_{pe}^2 \frac{\tilde{\psi}}{\alpha} \tilde{\mathbf{E}}_3 \\ \left(\frac{\partial^2}{\partial t^2} - c_s^2 \nabla^2 \right) \tilde{\psi} &= -\frac{\alpha q_e^2 k_1^2}{m_i m_e \omega_2 \omega_3} \langle \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3 \rangle. \end{aligned} \quad (15.83)$$

The replacement $\nabla^2 \langle \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3 \rangle \rightarrow -k_1^2 \langle \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3 \rangle$ has been made in the third line because, as stipulated in the discussion of Eq. (15.78), $\langle \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3 \rangle$ refers to the component of the product oscillating at the beat frequency $\omega_1 = \omega_3 - \omega_2$ and having the beat wavevector $\mathbf{k}_1 = \mathbf{k}_3 - \mathbf{k}_2$. Equating the coefficients of the nonlinear coupling terms in the second and third lines of Eq. (15.83) gives

$$\alpha = \omega_{pe} \frac{\sqrt{m_i m_e \omega_2 \omega_3}}{q_e k_1} \quad (15.84)$$

so the equations become

$$\begin{aligned} \left(\frac{\partial^2}{\partial t^2} + \omega_{pe}^2 - c^2 \nabla^2 \right) \tilde{\mathbf{E}}_3 &= -\lambda \tilde{\psi} \tilde{\mathbf{E}}_2 \\ \left(\frac{\partial^2}{\partial t^2} + \omega_{pe}^2 - \frac{3\kappa T_e}{m_e} \nabla^2 \right) \tilde{\mathbf{E}}_2 &= -\lambda \tilde{\psi} \tilde{\mathbf{E}}_3 \\ \left(\frac{\partial^2}{\partial t^2} - c_s^2 \nabla^2 \right) \tilde{\psi} &= -\lambda \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3, \end{aligned} \quad (15.85)$$

where

$$\lambda = \frac{\omega_{pe} q_e k_1}{\sqrt{m_i m_e \omega_2 \omega_3}} \quad (15.86)$$

and the angle brackets have now been omitted for clarity.

Defining the mode frequencies as

$$\begin{aligned}\omega_3^2 &= \omega_{pe}^2 + k_3^2 c^2 \\ \omega_2^2 &= \omega_{pe}^2 + 3k_2^2 \kappa T_e / m_e \\ \omega_1^2 &= k_1^2 c_s^2\end{aligned}\quad (15.87)$$

the coupled equations become

$$\begin{aligned}\left(\frac{\partial^2}{\partial t^2} + \omega_3^2\right) \tilde{\mathbf{E}}_3 &= -\lambda \tilde{\psi} \tilde{\mathbf{E}}_2 \\ \left(\frac{\partial^2}{\partial t^2} + \omega_2^2\right) \tilde{\mathbf{E}}_2 &= -\lambda \tilde{\psi} \tilde{\mathbf{E}}_3 \\ \left(\frac{\partial^2}{\partial t^2} + \omega_1^2\right) \tilde{\psi} &= -\lambda \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3,\end{aligned}\quad (15.88)$$

which is identical to the coupled oscillator system described by Eq. (15.3) if m is set to unity in Eq. (15.3) and $\tilde{\mathbf{E}}_2$ is parallel to $\tilde{\mathbf{E}}_3$. Using Eq. (15.33), the nonlinear growth rate is found to be

$$\begin{aligned}\gamma &= \frac{\lambda E_3}{4\sqrt{\omega_1 \omega_2}} \\ &= \frac{1}{4} \frac{\omega_{pi}}{\omega_2} \sqrt{\frac{\omega_3}{\omega_1}} k_1 \frac{q_e E_3}{m_e \omega_3}.\end{aligned}\quad (15.89)$$

15.4 Instability onset via nonlinear dispersion method

An equivalent way of considering the effect of nonlinearity is to derive a so-called nonlinear dispersion relation (Nishikawa 1968a, Nishikawa 1968b). This method has the virtue that both wave damping and frequency mismatches can easily be incorporated. To see how damping can be introduced, consider an electrical circuit consisting of an inductor, capacitor, and resistor all in series. The circuit equation is

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = 0, \quad (15.90)$$

where Q is the charge stored in the capacitor and the current is $I = dQ/dt$. The general solution is $Q \sim e^{-i\omega t}$, where ω satisfies

$$\omega^2 + i\omega \frac{R}{L} - \frac{1}{LC} = 0. \quad (15.91)$$

Solving for ω gives the usual damped harmonic oscillator solution

$$\omega = -\frac{iR}{2L} \pm \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}. \quad (15.92)$$

Thus, if $1/LC \gg R/L$ the imaginary part of the frequency is $\omega_i = -R/2L$ and the real part of the frequency is $\omega_r = \pm 1/\sqrt{LC}$. This means that the resistor can be identified with damping by the relation $R \rightarrow -2\omega_i L$. Interpreting Q_j now as the dependent variable for oscillation mode j , the appropriate differential equation for the damped mode can be written as

$$\frac{d^2 Q_j}{dt^2} + 2\Gamma_j \frac{dQ_j}{dt} + \omega_j^2 Q_j = 0, \quad (15.93)$$

where $\Gamma_j = |\text{Im } \omega_j|$ is the linear damping rate for mode j . Mode j therefore has the time dependence $\exp(\pm i\omega_j t - \Gamma_j t)$, where both ω_j and Γ_j are positive quantities.

Thus, when dissipation is included, the typical system given by Eq. (15.88) generalizes to

$$\begin{aligned} \left(\frac{\partial^2}{\partial t^2} + 2\Gamma_3 \frac{\partial}{\partial t} + \omega_3^2 \right) \tilde{\mathbf{E}}_3 &= -\lambda \tilde{\psi} \tilde{\mathbf{E}}_2 \\ \left(\frac{\partial^2}{\partial t^2} + 2\Gamma_2 \frac{\partial}{\partial t} + \omega_2^2 \right) \tilde{\mathbf{E}}_2 &= -\lambda \tilde{\psi} \tilde{\mathbf{E}}_3 \\ \left(\frac{\partial^2}{\partial t^2} + 2\Gamma_1 \frac{\partial}{\partial t} + \omega_1^2 \right) \tilde{\psi} &= -\lambda \tilde{\mathbf{E}}_2 \cdot \tilde{\mathbf{E}}_3. \end{aligned} \quad (15.94)$$

Let us consider once again the situation where the pump wave $\tilde{\mathbf{E}}_3$ is very large and so may be considered as having approximately constant amplitude. The decay waves then do not grow to sufficient amplitude to deplete the pump energy and so only the last two of the three coupled equations have to be considered. We define $\hat{s}_2$ as a unit vector in the direction of $\tilde{\mathbf{E}}_2$ so that

$$\tilde{\mathbf{E}}_2 = \hat{s}_2 \cdot \tilde{\mathbf{E}}_2 \quad (15.95)$$

and the last two of the three coupled equations become

$$\begin{aligned} \left(\frac{\partial^2}{\partial t^2} + 2\Gamma_2 \frac{\partial}{\partial t} + \omega_2^2 \right) \tilde{E}_2 &= -\lambda \tilde{\psi} \hat{s}_2 \cdot \tilde{\mathbf{E}}_3 \\ \left(\frac{\partial^2}{\partial t^2} + 2\Gamma_1 \frac{\partial}{\partial t} + \omega_1^2 \right) \tilde{\psi} &= -\lambda \tilde{E}_2 \hat{s}_2 \cdot \tilde{\mathbf{E}}_3. \end{aligned} \quad (15.96)$$

The wavevector selection rules are assumed to be satisfied but a slight mismatching in the frequency selection rules will be allowed. The pump wave is now written as

$$\hat{s}_2 \cdot \tilde{\mathbf{E}}_3 = Z_3 \cos \omega_3 t, \quad (15.97)$$

where Z_3 is the effective pump wave amplitude. The coupled daughter equations then become

$$\begin{aligned} \left(\frac{\partial^2}{\partial t^2} + 2\Gamma_2 \frac{\partial}{\partial t} + \omega_2^2 \right) \tilde{E}_2 &= -\frac{\lambda Z_3}{2} \left(e^{i\omega_3 t} + e^{-i\omega_3 t} \right) \tilde{\psi} \\ \left(\frac{\partial^2}{\partial t^2} + 2\Gamma_1 \frac{\partial}{\partial t} + \omega_1^2 \right) \tilde{\psi} &= -\frac{\lambda Z_3}{2} \left(e^{i\omega_3 t} + e^{-i\omega_3 t} \right) \tilde{E}_2. \end{aligned} \quad (15.98)$$

The Fourier transform of a quantity $f(t)$ is defined as

$$f(\omega) = \int dt f(t) e^{i\omega t}, \quad (15.99)$$

where the sign of the exponent is chosen to be consistent with the convention that a single Fourier mode has the form $\exp(-i\omega t)$. The Fourier transform of $f(t)e^{\pm i\omega_3 t}$ will therefore be

$$\int dt \left(f(t) e^{\pm i\omega_3 t} \right) e^{i\omega t} = f(\omega \pm \omega_3) \quad (15.100)$$

and the Fourier transform of $\partial f / \partial t$ will be

$$\int dt \left(\frac{\partial}{\partial t} f(t) \right) e^{i\omega t} = -i\omega \int dt f(t) e^{i\omega t} = -i\omega f(\omega). \quad (15.101)$$

Application of Eqs. (15.100) and (15.101) to Eqs. (15.98) gives

$$\begin{aligned} (-\omega^2 - 2i\omega\Gamma_2 + \omega_2^2) \tilde{E}_2(\omega) &= -\frac{\lambda Z_3}{2} \left[\tilde{\psi}(\omega + \omega_3) + \tilde{\psi}(\omega - \omega_3) \right] \\ (-\omega^2 - 2i\omega\Gamma_1 + \omega_1^2) \tilde{\psi}(\omega) &= \frac{\lambda Z_3}{2} \left[\tilde{E}_2(\omega + \omega_3) + \tilde{E}_2(\omega - \omega_3) \right]. \end{aligned} \quad (15.102)$$

By defining a generic linear dispersion relation

$$\varepsilon_j(\omega) = -\omega^2 - 2i\omega\Gamma_j + \omega_j^2, \quad (15.103)$$

these equations can be written as

$$\varepsilon_2(\omega) \tilde{E}_2(\omega) = -\frac{\lambda Z_3}{2} \left[\tilde{\psi}(\omega + \omega_3) + \tilde{\psi}(\omega - \omega_3) \right] \quad (15.104)$$

$$\varepsilon_1(\omega) \tilde{\psi}(\omega) = -\frac{\lambda Z_3}{2} \left[\tilde{E}_2(\omega + \omega_3) + \tilde{E}_2(\omega - \omega_3) \right]. \quad (15.105)$$

The frequency ω in Eq. (15.105) can be replaced by $\omega \pm \omega_3$ to give

$$\tilde{\psi}(\omega \pm \omega_3) = -\frac{\lambda Z_3}{2\varepsilon_1(\omega \pm \omega_3)} [\tilde{E}_2(\omega) + \tilde{E}_2(\omega \pm 2\omega_3)], \quad (15.106)$$

which is then substituted into Eq. (15.104) to obtain

$$\varepsilon_2(\omega)\tilde{E}_2(\omega) = \left(\frac{\lambda Z_3}{2}\right)^2 \left[\frac{\tilde{E}_2(\omega) + \tilde{E}_2(\omega + 2\omega_3)}{\varepsilon_1(\omega + \omega_3)} + \frac{\tilde{E}_2(\omega) + \tilde{E}_2(\omega - 2\omega_3)}{\varepsilon_1(\omega - \omega_3)} \right]. \quad (15.107)$$

The terms $\tilde{E}_2(\omega \pm 2\omega_3)$ can be discarded because, being non-resonant, they have insignificant amplitude. What remains is

$$\varepsilon_2(\omega) = \left(\frac{\lambda Z_3}{2}\right)^2 \left[\frac{1}{\varepsilon_1(\omega + \omega_3)} + \frac{1}{\varepsilon_1(\omega - \omega_3)} \right], \quad (15.108)$$

which is called the *nonlinear dispersion relation* (Nishikawa 1968a, Nishikawa 1968b).

The nonlinear dispersion relation is investigated by first writing

$$\omega = x + iy, \quad (15.109)$$

where it is assumed

$$x \simeq \omega_2 \quad (15.110)$$

and also

$$\omega - \omega_3 = x + iy - \omega_3 \simeq -\omega_1. \quad (15.111)$$

These assumptions and definitions have been made so that $\varepsilon_1(\omega - \omega_3)$ is close to zero, $\varepsilon_1(\omega + \omega_3)$ is not close to zero, and positive y corresponds to instability. The term $1/\varepsilon_1(\omega + \omega_3)$ can therefore be discarded as being non-resonant and the nonlinear dispersion relation simplifies to

$$\varepsilon_2(\omega)\varepsilon_1(\omega - \omega_3) = \left(\frac{\lambda Z_3}{2}\right)^2. \quad (15.112)$$

The linear dispersion relations ε_2 and ε_1 on the left-hand side are each Taylor-expanded to give

$$\begin{aligned} \varepsilon_2(\omega) &= \varepsilon_2(\omega_2 + x - \omega_2 + iy) \\ &\simeq \varepsilon_2(\omega_2) + (x - \omega_2 + iy) \left. \frac{d\varepsilon_2}{d\omega} \right|_{\omega=\omega_2} \\ &= -2i\omega_2\Gamma_2 - 2\omega_2(x - \omega_2 + iy) \end{aligned} \quad (15.113)$$

and

$$\begin{aligned}\varepsilon_1(\omega - \omega_3) &= \varepsilon_1(-\omega_1 + x + iy + \omega_1 - \omega_3) \\ &\simeq \varepsilon_2(-\omega_1) + (x + iy + \omega_1 - \omega_3) \left. \frac{d\varepsilon_1}{d\omega} \right|_{\omega=-\omega_1} \\ &= 2i\omega_1\Gamma_1 + 2\omega_1(x + iy + \omega_1 - \omega_3).\end{aligned}\quad (15.114)$$

Using these expansions, the nonlinear dispersion relation becomes

$$\{-i(\Gamma_2 + y) - (x - \omega_2)\} \{i(\Gamma_1 + y) + (x + \omega_1 - \omega_3)\} = \frac{1}{\omega_1\omega_2} \left(\frac{\lambda Z_3}{4} \right)^2 \quad (15.115)$$

or, in more compact form,

$$(\bar{x} + i(\Gamma_2 + y))(\bar{x} - \Delta + i(\Gamma_1 + y)) = -\frac{1}{\omega_1\omega_2} \left(\frac{\lambda Z_3}{4} \right)^2, \quad (15.116)$$

where

$$\bar{x} = x - \omega_2 \quad (15.117)$$

and

$$\Delta = \omega_3 - (\omega_1 + \omega_2) \quad (15.118)$$

is the frequency mismatch.

The real and imaginary parts of Eq. (15.116) are

$$(\bar{x} - \Delta)\bar{x} - (\Gamma_2 + y)(\Gamma_1 + y) = -\frac{1}{\omega_1\omega_2} \left(\frac{\lambda Z_3}{4} \right)^2 \quad (15.119)$$

$$\bar{x}(\Gamma_1 + y) + (\bar{x} - \Delta)(\Gamma_2 + y) = 0. \quad (15.120)$$

Solving Eq. (15.120) gives

$$\bar{x} = \frac{(\Gamma_2 + y)}{(2y + \Gamma_1 + \Gamma_2)}\Delta, \quad \bar{x} - \Delta = -\frac{(\Gamma_1 + y)}{(2y + \Gamma_1 + \Gamma_2)}\Delta \quad (15.121)$$

and substitution for $\tilde{x}$ and $\tilde{x} - \Delta$ in Eq. (15.119) results in the expression for the growth rate

$$(\Gamma_2 + y)(\Gamma_1 + y) \left(1 + \frac{\Delta^2}{(2y + \Gamma_1 + \Gamma_2)^2} \right) = \frac{1}{\omega_1 \omega_2} \left(\frac{\lambda Z_3}{4} \right)^2. \quad (15.122)$$

Onset of instability corresponds to y being close to zero and at marginal stability Eq. (15.122) becomes

$$(\Gamma_2 + \Gamma_1)y + \Gamma_1 \Gamma_2 = \frac{\lambda^2 Z_3^2}{16\omega_1 \omega_2 \left(1 + \frac{\Delta^2}{(\Gamma_2 + \Gamma_1)^2} \right)}. \quad (15.123)$$

The growth rate near threshold thus is

$$y = \frac{1}{(\Gamma_2 + \Gamma_1)} \left(\frac{\lambda^2 Z_3^2}{16\omega_1 \omega_2 \left(1 + \frac{\Delta^2}{(\Gamma_2 + \Gamma_1)^2} \right)} - \Gamma_1 \Gamma_2 \right). \quad (15.124)$$

This shows that the pump amplitude threshold for instability is

$$Z_3 = \frac{4\sqrt{\omega_1 \omega_2 \Gamma_1 \Gamma_2}}{\lambda} \sqrt{1 + \frac{\Delta^2}{(\Gamma_2 + \Gamma_1)^2}}; \quad (15.125)$$

this threshold is proportional to the geometric mean of the linear damping rates of the two modes. If the pump amplitude is below threshold then the nonlinear instability does not occur. The lowest threshold occurs when $\Delta = 0$, i.e., when the frequency selection rule is exactly satisfied. Instability when the pump amplitude exceeds a threshold is routinely observed in actual experimental situations as was discussed earlier, i.e., decay instability of a pump wave is observed to begin only when the pump wave amplitude exceeds threshold (for example, see Stenzel and Wong (1972)). For a pump amplitude well above threshold, Eq. (15.122) becomes

$$y = \sqrt{\frac{1}{\omega_1 \omega_2} \left(\frac{\lambda Z_3}{4} \right)^2 - \frac{\Delta^2}{4}}, \quad (15.126)$$

which shows that frequency mismatch reduces the growth rate. The situation reduces to Eqs. (15.33) and (15.89) when $\Delta = 0$. Thus the nonlinear dispersion relation formalism extends the Manley-Rowe coupled oscillator model to include the effects of both dissipation and frequency mismatch.

15.5 Digging a density hole via ponderomotive force

If the low-frequency daughter mode is at zero frequency, then the ion acoustic mode ceases to be a wave. Instead, it becomes a density depletion caused by the ponderomotive force and so Eq. (15.56) reduces to

$$\frac{1}{n} \nabla \tilde{n} = -\frac{1}{2} \frac{m_e}{\kappa T_e} \nabla (\tilde{u}_e^h)^2, \quad (15.127)$$

which can be integrated to give a Boltzmann-like relation

$$\frac{\tilde{n}}{n} = -\frac{1}{2} \frac{m_e \langle (\tilde{u}_e^h)^2 \rangle}{\kappa T_e}. \quad (15.128)$$

Since the high-frequency daughter wave is the same as the pump wave there is now only one high-frequency wave and so there is no need to have subscripts distinguishing the modes. The consequence is that the high-frequency wave propagates in a plasma having a density depletion dug out by the ponderomotive force associated with the wave. For example, the Langmuir wave equation in this case would become

$$\left(\frac{\partial^2}{\partial t^2} + 2\Gamma \frac{\partial}{\partial t} + \omega_{pe}^2 \left(1 - \frac{1}{2} \frac{m_e \langle (\tilde{u}_e^h)^2 \rangle}{\kappa T_e} \right) - \frac{3\kappa T_e}{m_e} \nabla^2 \right) \tilde{\mathbf{u}}_e^h = 0, \quad (15.129)$$

where $\tilde{\mathbf{u}}_e^h$ has been used as the linear variable instead of $\tilde{\mathbf{E}}$ and a linear damping term involving the linear damping rate Γ has been introduced.

An undamped linear Langmuir wave in a uniform plasma satisfies the dispersion relation $\omega^2 = \omega_{pe}^2 (1 + 3k^2 \lambda_{De}^2)$, where $k\lambda_{De} \ll 1$ so that the wave frequency is very close to ω_{pe} . It is reasonable to presume that the nonlinear wave also oscillates at a frequency very close to ω_{pe} and so what is important is the deviation of the frequency from ω_{pe} . To investigate this, the electron fluid velocity is assumed to be of the form

$$\begin{aligned} \tilde{\mathbf{u}}_e^h(\mathbf{x}, t) &= \text{Re} \left[\mathbf{A}(x, t) e^{-i\omega_{pe}t} \right] \\ &= \frac{1}{2} \left\{ \mathbf{A}(x, t) e^{-i\omega_{pe}t} + \mathbf{A}^*(x, t) e^{i\omega_{pe}t} \right\}, \end{aligned} \quad (15.130)$$

in which case

$$\left\langle (\tilde{\mathbf{u}}_e^h)^2 \right\rangle = \frac{1}{2} |\mathbf{A}|^2 \quad (15.131)$$

and the time dependence of $\mathbf{A}(x, t)$ characterizes the extent to which the wave frequency deviates from ω_{pe} . Because this deviation is small, $\mathbf{A}$ changes slowly compared to ω_{pe} , and so in analogy to Eq. (3.22) it is possible to approximate

$$\frac{\partial^2}{\partial t^2} [\mathbf{A}(x, t)e^{-i\omega_{pe}t}] \simeq -\omega_{pe}^2 \mathbf{A}(x, t)e^{-i\omega_{pe}t} - 2i\omega_{pe} \frac{\partial \mathbf{A}}{\partial t} e^{-i\omega_{pe}t} \quad (15.132)$$

and

$$2\Gamma \frac{\partial}{\partial t} [\mathbf{A}(x, t)e^{-i\omega_{pe}t}] \approx -i2\omega_{pe}\Gamma \mathbf{A}e^{-i\omega_{pe}t}.$$

Substitution of Eq. (15.132) into Eq. (15.129) gives

$$2i\omega_{pe} \frac{\partial \mathbf{A}}{\partial t} + i2\omega_{pe}\Gamma \mathbf{A} + \frac{\omega_{pe}^2}{4} \frac{m_e}{\kappa T_e} |\mathbf{A}|^2 \mathbf{A} + c^2 \nabla^2 \mathbf{A} = 0. \quad (15.133)$$

By defining the normalized variables

$$\begin{aligned} \tau &= \omega_{pe}t/2 \\ \chi &= \frac{\mathbf{A}}{2\sqrt{\kappa T_e/m_e}} \\ \xi &= \mathbf{x}\omega_{pe}/c \\ \eta &= 2\Gamma/\omega_{pe} \end{aligned} \quad (15.134)$$

Eq. (15.133) can be put in the standardized form

$$i\frac{\partial \chi}{\partial \tau} + i\eta\chi + |\chi|^2 \chi + \nabla_\xi^2 \chi = 0; \quad (15.135)$$

this is called a nonlinear Schrödinger equation since, if $\eta = 0$, this equation resembles a Schrödinger equation where $|\chi|^2$ plays the role of a potential energy.

In order to exploit this analogy, we recall the relationship between the Schrödinger equation and the classical conservation of energy relation for a particle in a potential well $V(x)$. According to classical mechanics, the sum of the kinetic and potential energies gives the total energy, i.e.,

$$\frac{p^2}{2m} + V(\mathbf{x}) = E. \quad (15.136)$$

However, in quantum mechanics, the momentum and the energy are expressed as spatial and temporal operators, $\mathbf{p} = -i\hbar\nabla$ and $E = i\hbar\partial/\partial t$, which act on a wave function ψ so that Eq. (15.136) becomes

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = i\hbar \frac{\partial \psi}{\partial t} \quad (15.137)$$

or, after rearrangement,

$$i\hbar \frac{\partial \psi}{\partial t} - V\psi + \frac{\hbar^2}{2m} \nabla^2 \psi = 0. \quad (15.138)$$

Equation (15.136) shows that a particle will be trapped in a potential well if $V(\pm\infty) > E > V_{\min}$, where $V_{\min}$ is the minimum value of V . From the quantum mechanical point of view, $|\psi|^2$ is the probability of finding the particle at position $\mathbf{x}$. Thus, existence of solutions to Eq. (15.138) localized to the vicinity of $V_{\min}$ is the quantum mechanical way of stating that a particle can be trapped in a potential well. Comparison of Eqs. (15.135) and (15.138) shows that $-|\bar{\mathbf{A}}|^2$ plays the role of V and so a local maximum of $|\bar{\mathbf{A}}|^2$ should act as an effective potential well. This makes physical sense because Langmuir waves reflect from regions of high density and the amplitude-dependent ponderomotive force digs a hole in the plasma. Thus, regions of high wave amplitude create a density depression and the Langmuir wave reflects from the high-density regions surrounding this density depression. The Langmuir wave then becomes trapped in a depression of its own making. Formation of this depression can be an unstable process because if a wave is initially trapped in a shallow well, its energy $|\chi|^2$ will concentrate at the bottom of this well, but this concentration of $|\chi|^2$ will make the well deeper and so concentrate the wave energy into a smaller region, making $|\chi|^2$ even larger, and so on.

Caviton instability

The instability outlined above can be described in a quantitative manner using the 1-D version of Eq. (15.135), namely

$$i \frac{\partial \chi}{\partial \tau} + i\eta\chi + |\chi|^2 \chi + \frac{\partial^2 \chi}{\partial \xi^2} = 0. \quad (15.139)$$

It is assumed that a stable solution $\chi_0(x, t)$ exists initially and satisfies

$$i \frac{\partial \chi_0}{\partial \tau} + i\eta\chi_0 + |\chi_0|^2 \chi_0 + \frac{\partial^2 \chi_0}{\partial \xi^2} = 0, \quad (15.140)$$

where $|\chi_0(x, t)|$ is bounded in both time and space. Next, a slightly different solution is considered,

$$\chi(x, t) = \chi_0(x, t) + \tilde{\chi}(x, t), \quad (15.141)$$

where the perturbation $\tilde{\chi}(x, t)$ is assumed to be small compared to $\chi_0(x, t)$. The equation for $\chi(x, t)$ is thus

$$i \frac{\partial}{\partial \tau} (\chi_0 + \tilde{\chi}) + i\eta(\chi_0 + \tilde{\chi}) + |\chi_0 + \tilde{\chi}|^2 (\chi_0 + \tilde{\chi}) + \frac{\partial^2}{\partial \xi^2} (\chi_0 + \tilde{\chi}) = 0. \quad (15.142)$$

Subtracting Eq. (15.140) from (15.142) yields

$$i \frac{\partial \tilde{\chi}}{\partial \tau} + i \eta \tilde{\chi} + |\chi_0 + \tilde{\chi}|^2 (\chi_0 + \tilde{\chi}) - |\chi_0|^2 \chi_0 + \frac{\partial^2 \tilde{\chi}}{\partial \xi^2} = 0. \quad (15.143)$$

Expansion of the potential-energy-like terms while keeping only terms linear in the perturbation gives

$$|\chi_0 + \tilde{\chi}|^2 (\chi_0 + \tilde{\chi}) - |\chi_0|^2 \chi_0 \approx \chi_0^2 \tilde{\chi}^* + 2 |\chi_0|^2 \tilde{\chi} \quad (15.144)$$

so Eq. (15.143) becomes

$$i \frac{\partial \tilde{\chi}}{\partial \tau} + i \eta \tilde{\chi} + \chi_0^2 \tilde{\chi}^* + 2 |\chi_0|^2 \tilde{\chi} + \frac{\partial^2 \tilde{\chi}}{\partial \xi^2} = 0. \quad (15.145)$$

It is now assumed that the perturbation is unstable and has the space-time dependence

$$\tilde{\chi} \sim e^{ik\xi + \gamma t}, \quad (15.146)$$

in which case Eq. (15.145) becomes

$$(i\gamma + i\eta + 2 |\chi_0|^2 - k^2) \tilde{\chi} = -\chi_0^2 \tilde{\chi}^*, \quad (15.147)$$

which has the complex conjugate

$$(-i\gamma - i\eta + 2 |\chi_0|^2 - k^2) \tilde{\chi}^* = -\chi_0^{*2} \tilde{\chi}. \quad (15.148)$$

Combining the above two equations gives a dispersion relation for the growth rate

$$(\gamma + \eta)^2 = -k^4 + 4k^2 |\chi_0|^2 - 3 |\chi_0|^4. \quad (15.149)$$

The maximum γ is found by taking the derivative of both sides with respect to k^2 and setting this derivative to zero, obtaining

$$k_{\max}^2 = 2 |\chi_0|^2 \quad (15.150)$$

as the value of k^2 giving the maximum for γ . Substitution of $k_{\max}^2$ into Eq. (15.149) gives

$$\gamma_{\max} = -\eta + |\chi_0|^2. \quad (15.151)$$

Thus, the configuration is unstable when $|\chi_0|^2 > \eta$ or, in terms of the original variables, when

$$\frac{m_e \left\langle (\tilde{u}_e^h)^2 \right\rangle}{4\kappa T_e} > \frac{\Gamma}{\omega_{pe}}. \quad (15.152)$$

This is called a caviton instability and it tends to dig a sharp hole in the plasma density. This is because each successive stage of growth can be considered a

quasi-equilibrium, which is destabilized, and from Eq. (15.150) it is seen that the most unstable k becomes progressively larger as the amplitude increases. Another way of seeing this hole-digging tendency is to note that Eqs. (15.147) and (15.148) are coupled by their respective right-hand terms and these terms are proportional to the amplitude of the original wave. It is this coupling that leads to instability of the perturbation and so the perturbation is most unstable where the amplitude of the original wave was largest. The digging of a density cavity in a plasma by a Langmuir wave was observed experimentally by Kim, Stenzel and Wong (1974); the density cavity was called a caviton.

Stationary Envelope Soliton

A special, fully nonlinear solution of Eq. (15.139) can be found in the limit where damping is sufficiently small to be neglected so that the nonlinear Schrödinger equation reduces to

$$i \frac{\partial \chi}{\partial \tau} + |\chi|^2 \chi + \frac{\partial^2 \chi}{\partial \xi^2} = 0. \quad (15.153)$$

We now search for a solution that vanishes at infinity, propagates at some fixed velocity, and oscillates so that

$$\chi = g(\xi) e^{i\Omega\tau}. \quad (15.154)$$

With this assumption, Eq. (15.153) becomes

$$-\Omega g + g^3 + g'' = 0. \quad (15.155)$$

After multiplying by the integrating factor g' , this becomes

$$\frac{d}{d\xi} \left(-\frac{\Omega}{2} g^2 + \frac{1}{4} g^4 + \frac{1}{2} (g')^2 \right) = 0. \quad (15.156)$$

Since the solution is assumed to vanish at infinity, integration with respect to ξ from $\xi = -\infty$ gives

$$(g')^2 = \Omega g^2 - \frac{1}{2} g^4 \quad (15.157)$$

or

$$\frac{dg}{g\sqrt{\Omega - \frac{1}{2}g^2}} = d\xi. \quad (15.158)$$

By letting

$$g = \frac{\sqrt{2\Omega}}{\cosh \theta} \quad (15.159)$$

it is seen that

$$dg = -\frac{\sqrt{2\Omega}}{\cosh^2 \theta} \sinh \theta d\theta, \quad (15.160)$$

in which case

$$\theta = -\sqrt{\Omega}\xi + \delta, \quad (15.161)$$

where δ is an arbitrary constant. Thus, the solution is

$$\chi(\xi, \tau) = \frac{\sqrt{2\Omega}e^{i\Omega\tau}}{\cosh(\sqrt{\Omega}\xi - \delta)}, \quad (15.162)$$

which is localized in space. For large Ω , the oscillation frequency and amplitude both increase, and the localization is more pronounced.

Propagating envelope soliton

One may ask whether the above solution can be generalized to a propagating solution, i.e., can ξ be replaced by $\xi - vt$, where v is a velocity? Making only this replacement is clearly inadequate and so a solution of the form

$$\chi = g(\xi - vt)e^{i\Omega\tau + ih(\xi, \tau)} \quad (15.163)$$

is assumed, where $h(\xi, \tau)$ is an unknown function to be determined. Substitution of this assumed solution into Eq. (15.153) gives

$$-ivg' - \Omega g - g\frac{\partial h}{\partial \tau} + g^3 + g'' + 2i\frac{\partial h}{\partial \xi}g' - \left(\frac{\partial h}{\partial \xi}\right)^2 g = 0. \quad (15.164)$$

Setting the imaginary part to zero gives

$$v = 2\frac{\partial h}{\partial \xi}, \quad (15.165)$$

which can be integrated to give

$$h = \frac{v\xi}{2} + f(\tau), \quad (15.166)$$

where $f(\tau)$ is to be determined. The real part of the equation becomes

$$-\Omega g - g\frac{\partial h}{\partial \tau} + g^3 + g'' - \frac{v^2}{4}g = 0. \quad (15.167)$$

If we set

$$\frac{\partial h}{\partial \tau} = -\frac{v^2}{4} \quad (15.168)$$

then the second and last terms in Eq. (15.167) cancel each other. Equation (15.168) implies $f(\tau) = -v^2\tau/4$ so

$$h(\xi, \tau) = \frac{v\xi}{2} - \frac{v^2}{4}\tau. \quad (15.169)$$

Thus, Eq. (15.167) reverts to Eq. (15.155) which is solved as in Eqs. (15.156)–(15.162) to give the propagating envelope soliton

$$\chi(\xi, \tau) = \frac{\sqrt{2\Omega} \exp(i\Omega\tau + iv\xi/2 - iv^2\tau/4)}{\cosh\left(\sqrt{\Omega}(\xi - v\tau) - \delta\right)}. \quad (15.170)$$

15.6 Ion acoustic wave soliton

The ion acoustic wave dispersion relation is

$$\omega^2 = \frac{k^2 c_s^2}{1 + k^2 \lambda_D^2}, \quad (15.171)$$

which has the forward propagating solution

$$\omega = kc_s (1 + k^2 \lambda_D^2)^{-1/2} \quad (15.172)$$

or, for small $k\lambda_D$,

$$\omega = kc_s - k^3 \lambda_D^2 c_s / 2, \quad (15.173)$$

where the last term is small. Since $\partial/\partial x \rightarrow ik$ and $\partial/\partial t \rightarrow -i\omega$ the inverse substitution is $k \rightarrow -i\partial/\partial x$ and $\omega \rightarrow i\partial/\partial t$. Applying this inverse substitution to Eq. (15.173) shows that the forward propagating ion acoustic wave implies that the partial differential equation for, say, ion velocity would be

$$i \frac{\partial u_i}{\partial t} = -i \frac{\partial u_i}{\partial x} c_s - (-i)^3 \frac{\partial^3 u_i}{\partial x^3} \frac{\lambda_D^2 c_s}{2}. \quad (15.174)$$

After multiplying by $-i$, this gives the dispersive forward propagating wave equation

$$\frac{\partial u_i}{\partial t} = -c_s \frac{\partial u_i}{\partial x} - \frac{\lambda_D^2 c_s}{2} \frac{\partial^3 u_i}{\partial x^3}. \quad (15.175)$$

This wave equation was derived using a linearized version of the ion fluid equation of motion, namely

$$m_i \frac{\partial u_i}{\partial t} = q_i E. \quad (15.176)$$

If, instead, the complete nonlinear ion equation had been used, the ion fluid equation of motion would contain a convective nonlinear term and be

$$m_i \left(\frac{\partial u_i}{\partial t} + u_i \frac{\partial u_i}{\partial x} \right) = q_i E. \quad (15.177)$$

This suggests that inclusion of convective ion nonlinearity corresponds to making the generalization

$$\frac{\partial u_i}{\partial t} \rightarrow \frac{\partial u_i}{\partial t} + u_i \frac{\partial u_i}{\partial x} \quad (15.178)$$

and so the forward propagating ion acoustic wave equation with inclusion of ion convective nonlinearity is

$$\frac{\partial u_i}{\partial t} + (u_i + c_s) \frac{\partial u_i}{\partial x} + \frac{\lambda_D^2 c_s}{2} \frac{\partial^3 u_i}{\partial x^3} = 0. \quad (15.179)$$

This suggests defining a new variable

$$U = u_i + c_s, \quad (15.180)$$

which differs only by a constant from the ion fluid velocity. The forward propagating nonlinear ion acoustic wave equation can thus be rewritten as

$$\frac{\partial U}{\partial t} + U \frac{\partial U}{\partial x} + \frac{\lambda_D^2 c_s}{2} \frac{\partial^3 U}{\partial x^3} = 0. \quad (15.181)$$

We now introduce dimensionless variables by normalizing lengths to the Debye length, velocities to c_s , and time to $\omega_{pi} = c_s/\lambda_{de}$. The dimensionless variables are thus

$$\chi = \frac{U}{c_s}, \quad \xi = \frac{x}{\lambda_D}, \quad \tau = \omega_{pi} t, \quad (15.182)$$

and the dimensionless wave equation becomes

$$\frac{\partial \chi}{\partial \tau} + \chi \frac{\partial \chi}{\partial \xi} + \frac{1}{2} \frac{\partial^3 \chi}{\partial \xi^3} = 0; \quad (15.183)$$

this is called the Korteweg–de Vries (KdV) equation (Korteweg and de Vries 1895). Modern interest in this equation was stimulated with the discovery of a general solution to the soliton problem by Gardner *et al.* (1967).

Because the linear wave is forward traveling with unity velocity in these dimensionless variables, it is reasonable to postulate that the nonlinear solution has the forward propagating form

$$\chi = \chi(\xi - V\tau), \quad (15.184)$$

where V is of order unity. A special solution can be found by introducing the wave-frame position variable

$$\eta = \xi - V\tau \quad (15.185)$$

so the lab-frame space and time derivatives can be written as

$$\begin{aligned} \frac{\partial}{\partial \xi} &= \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial \tau} &= -V \frac{\partial}{\partial \eta}. \end{aligned} \quad (15.186)$$

Substitution of these into the wave equation gives an ordinary differential equation in the wave-frame,

$$-V \frac{d\chi}{d\eta} + \frac{1}{2} \frac{d\chi^2}{d\eta} + \frac{1}{2} \frac{d^3\chi}{d\eta^3} = 0, \quad (15.187)$$

where d has been used instead of ∂ because the equation is now an ordinary differential equation. A solution is sought that vanishes at both plus and minus infinity; such a solution is called a solitary wave. To find this solution, Eq. (15.187) is integrated using the boundary condition that χ vanishes at infinity to obtain

$$-V\chi + \frac{\chi^2}{2} + \frac{1}{2} \frac{d^2\chi}{d\eta^2} = 0. \quad (15.188)$$

Multiplying by the integration factor $d\chi/d\eta$ allows this to be recast as

$$\frac{d}{d\eta} \left(-\frac{V\chi^2}{2} + \frac{\chi^3}{6} + \frac{1}{4} \left(\frac{d\chi}{d\eta} \right)^2 \right) = 0 \quad (15.189)$$

and then integrating gives

$$\frac{d\chi}{d\eta} = \chi \sqrt{2V - \frac{2}{3}\chi}, \quad (15.190)$$

where the boundary condition at infinity has been used again. This can be written as

$$\frac{d(\chi/3)}{2^{1/2} \frac{\chi}{3} \sqrt{V - \frac{\chi}{3}}} = d\eta. \quad (15.191)$$

The substitution

$$\frac{\chi}{3} = \frac{V}{\cosh^2 \theta} \quad (15.192)$$

allows simplification of Eq. (15.191) to

$$d\left(\frac{\chi}{3}\right) = -\frac{2V \sinh \theta}{\cosh^3 \theta} d\theta. \quad (15.193)$$

Thus, Eq. (15.191) becomes

$$\frac{-\frac{2V \sinh \theta}{\cosh^3 \theta} d\theta}{2^{1/2} \frac{V}{\cosh^2 \theta} \sqrt{V - \frac{V}{\cosh^2 \theta}}} = d\eta \quad (15.194)$$

or

$$-\sqrt{\frac{2}{V}} d\theta = d\eta,$$

which can be integrated to give

$$\theta = \sqrt{\frac{V}{2}} (\xi_0 - \eta), \quad (15.195)$$

where ξ_0 is a constant. The propagating solitary wave solution to Eq. (15.183) is therefore

$$\chi(\eta) = \frac{3V}{\cosh^2 \left(\sqrt{\frac{V}{2}} (\xi_0 - \eta) \right)} \quad (15.196)$$

or

$$\chi(\xi, \tau) = \frac{3V}{\cosh^2 \left(\sqrt{\frac{V}{2}} (\xi_0 - (\xi - V\tau)) \right)}. \quad (15.197)$$

This solution, called a soliton, has the following properties:

1. As required, the solution vanishes when $|\xi| \rightarrow \infty$.
2. The spatial profile consists of a solitary pulse centered around the position $\xi = \xi_0 + V\tau$.
3. The pulse width scales as $V^{-1/2}$ and the pulse height scales as V , so larger amplitude solitons are sharper and propagate faster.

An important property of solitons is that they obey a form of superposition principle even though they are essentially nonlinear. In particular, when a fast soliton overtakes a slow soliton, it is seen that after the collision or interaction, both the fast and slow solitons retain their identity. The underlying mathematical theory explaining this surprising behavior is complex and beyond the scope of this text (the interested reader should consult Drazin and Johns (1989)). This theory

is called inverse scattering (Gardner *et al.* 1967) and involves mapping the nonlinear equation to a special linear equation, which is then solved and then mapped back again to give the time evolution of the nonlinear solution. The special linear equation describes a quantum mechanical particle trapped in a potential well, where $\chi(\xi, t)$ plays the role of the potential. In particular, the linear equation is of the form $\partial^2 \psi / \partial \xi^2 + (\lambda - \chi(\xi, t)) \psi = 0$, where λ is an eigenvalue and time t is treated as a parameter. When this linear equation is solved for $\chi(\xi, t)$ and the solution is inserted in the KdV equation it is found that $\partial \lambda / \partial t = 0$. This invariance of the eigenvalues is a key feature, which makes possible the {nonlinear $\rightarrow$ linear} and then {linear $\rightarrow$ nonlinear} mappings required to construct the solution.

15.7 Assignments

1. Pump depletion for the system of three coupled oscillators discussed in Section 15.2.
 - (a) Suppose $A_1 = 0$ at $t = 0$ but A_2 and A_3 are finite. What is the value of the constant in Eq. (15.27)?
 - (b) Suppose that $\sin \theta$ is not zero. Will A_1 become finite at times $t > 0$? If A_1 becomes finite, what constraint does Eq. (15.27) put on the value of $\cos \theta$ and hence on $\sin \theta$?
 - (c) Use Eqs. (15.24) to write $A_1^2(t)$ and $A_2^2(t)$ in terms of $A_3^2(t)$ and the initial conditions $A_0^2(0)$, $A_1^2(0)$, and $A_3^2(0)$.
 - (d) Square both sides of Eq. (15.18c) and use the results in (b) and (c) above to obtain an equation of the form

$$\frac{1}{2} \left(\frac{dA_3}{dt} \right)^2 + U(A_3) = E, \quad (15.198)$$

where E is a constant. Sketch the dependence of $U(A_3)$ on A_3 indicating the locations of maxima and minima.

- (e) Consider Eq. (15.198) as the energy equation for a pseudo-particle with velocity dA_3/dt in a potential well $U(A_3)$. What is the total energy of this pseudo-particle in terms of $A_0^2(0)$, $A_1^2(0)$, and $A_3^2(0)$? To what does the position of the pseudo-particle in the potential well correspond?
- (f) Considering the initial conditions given in (a), where is the pseudo-particle initially located in its potential well? Sketch the qualitative time dependence of A_3 and indicate how features of this time dependence correspond to the location of the pseudo-particle in its potential well.
- (g) Give an integral expression for the time required for A_3 to go to its first zero.
- (h) Solve Eqs. (15.18a)–(15.18c) numerically for the initial conditions given in (a) and use the numerical results to show that the time dependence of A_3 can be interpreted as the position of a pseudo-particle in a potential well prescribed by Eq. (15.198). Note: the reduction of the pump wave amplitude as it transfers energy to the daughter waves is called pump depletion; the mathematical behavior discussed in

this problem can be expressed in terms of Jacobi elliptic functions (Sagdeev and Galeev 1969).

2. Stimulated Raman scattering in laser fusion. Laser fusion is a proposed method for attaining a controlled fusion reaction. This method involves illuminating a millimeter radius pellet with intense laser light so as to ablate the outer layer of the pellet. The radial outflow of the ablating material constitutes a radial outflow of momentum and so, in order to conserve radial momentum, the rest of the pellet accelerates inwards in rocket-like fashion. An important issue is the decay of the incident laser light into other waves because such a decay would reduce the power available to drive the ablation process. For example, there could be stimulated Raman scattering where the incident laser light decays into an outward propagating electromagnetic wave (often called backscattered light) and an inward propagating Langmuir wave.
 - (a) Let the incident laser beam be denoted by subscript 3, the outward propagating electromagnetic wave be denoted by subscript 2, and the inward propagating Langmuir wave be denoted by subscript 1. Assume the laser wavelength is much shorter than the characteristic density scale length. Show that if the backward scattered electromagnetic wave has a frequency only slightly above ω_{pe} then it is possible to satisfy the frequency and wavenumber matching conditions at a location where the density is approximately 1/4 of the density at which the incident wave would reflect.
 - (b) Draw a sketch of ω versus k for the incident em wave, the backscattered em wave, and the Langmuir wave. Note that $v_{Te} \ll c$ so if this plot is scaled to show the dispersion of the electromagnetic waves, the Langmuir wave dispersion is almost a horizontal line.
 - (c) Draw vectors on the sketch in (a) with coordinates $\{k, \omega\}$ so the incident electromagnetic wave is a vector $\mathbf{v}_3 = \{k_3, \omega_3\}$, the backscattered wave is a vector $\mathbf{v}_2 = \{k_2, \omega_2\}$, and the Langmuir wave is a vector $\mathbf{v}_1 = \{k_1, \omega_1\}$. Show on the sketch how the vectors can add up in a manner consistent with the selection rules $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$.
3. Parametric decay instability. The minimum pump amplitude for the parametric decay instability (electromagnetic wave decays into a Langmuir wave and an ion acoustic wave) is given by Eq. (15.125) to be

$$E_3 = 4 \frac{\sqrt{\omega_1 \omega_2 \Gamma_1 \Gamma_2}}{\lambda},$$

where the coupling parameter was defined in Eq. (15.86) to be

$$\lambda = \frac{\omega_{pe} q k_1}{\sqrt{m_i m_e \omega_2 \omega_3}}.$$

- (a) How do the Langmuir wave frequency ω_2 and the ion acoustic wave frequency ω_1 compare to ω_{pe} (nearly same, much larger, or much smaller)?

- (b) Taking into account the selection rules, how does the pump frequency ω_3 compare to ω_{pe} ? What does this imply for k_3 and hence k_1 and k_2 ?
- (c) Using the results from (a) express the instability threshold in a form where $\varepsilon_0 E^2 / n \kappa T_e$ is a function of Γ_1 / ω_1 and Γ_2 / ω_2 .
- (d) Suppose a pump wave with frequency ω_3 propagates into a plasma with a monotonically increasing density. Show the relative locations of stimulated Brillouin and stimulated Raman backscattering in terms of a criterion for the local density. Assume the wavelengths of all modes are much shorter than the density gradient scale length.
- (e) The National Ignition Facility at the Lawrence Livermore National Lab in the USA has an array of 351 nm wavelength lasers designed to apply 5×10^{14} W over the surface of a target having an initial radius of 1 mm. If these lasers are considered to pump electromagnetic waves with frequency ω_3 , at what density does $\omega_3 = \omega_{pe}$ occur? Assuming $T_e = 300$ eV, what is the value of $\varepsilon_0 E^2 / n \kappa T_e$ at the location where $\omega_3 = \omega_{pe}$? If $\Gamma_1 / \omega_1 = \Gamma_2 / \omega_2 = 0.1$ will this plasma be susceptible to the parametric decay instability?